

**Application #7: A case study to explore
neighborhoods in kidney health and disease.**
Michael Eadon, MD

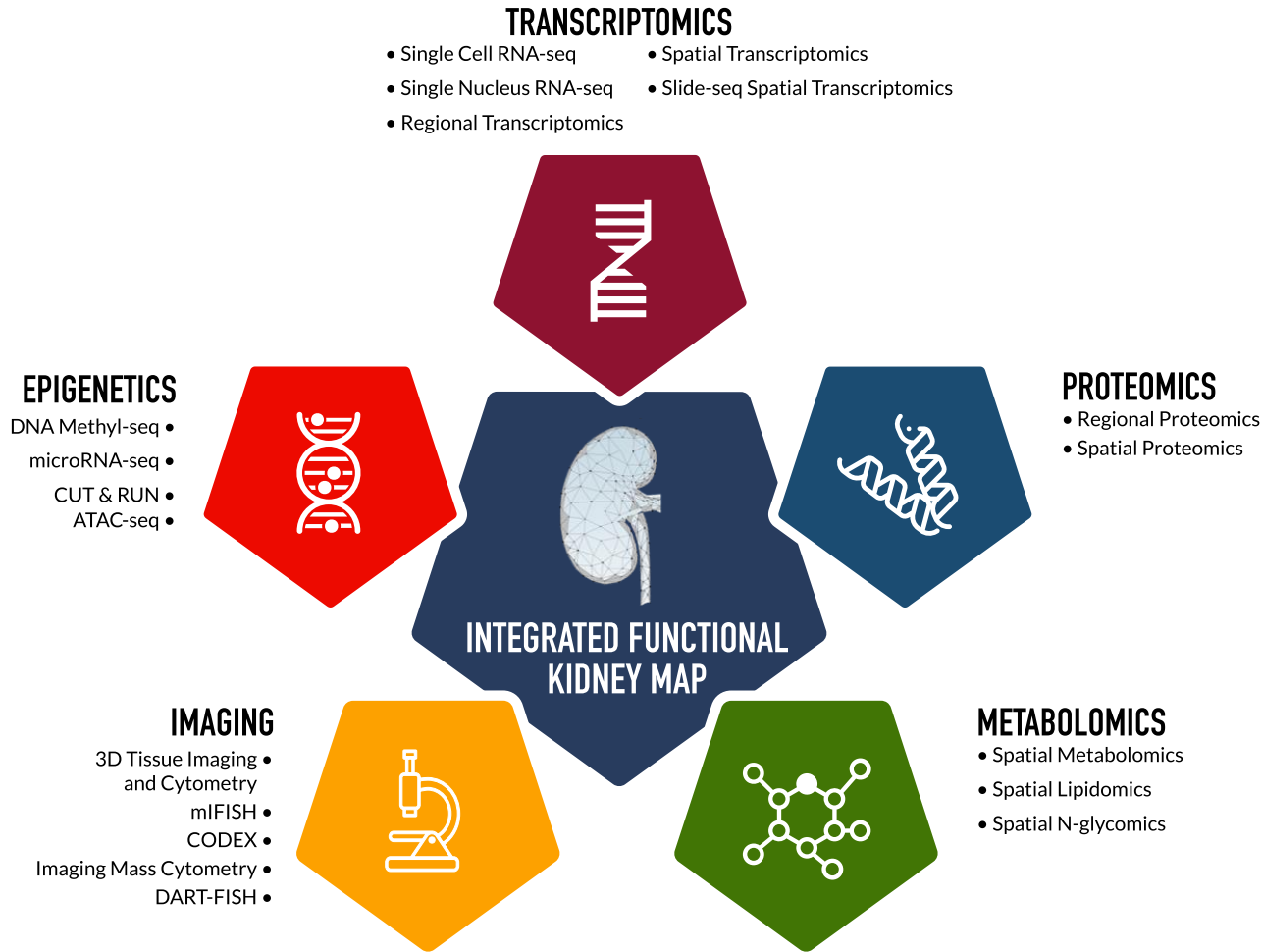


INDIANA UNIVERSITY
SCHOOL OF MEDICINE



Disclosures

- NIH Funding



Within the KPMP data repository, our team contributed:

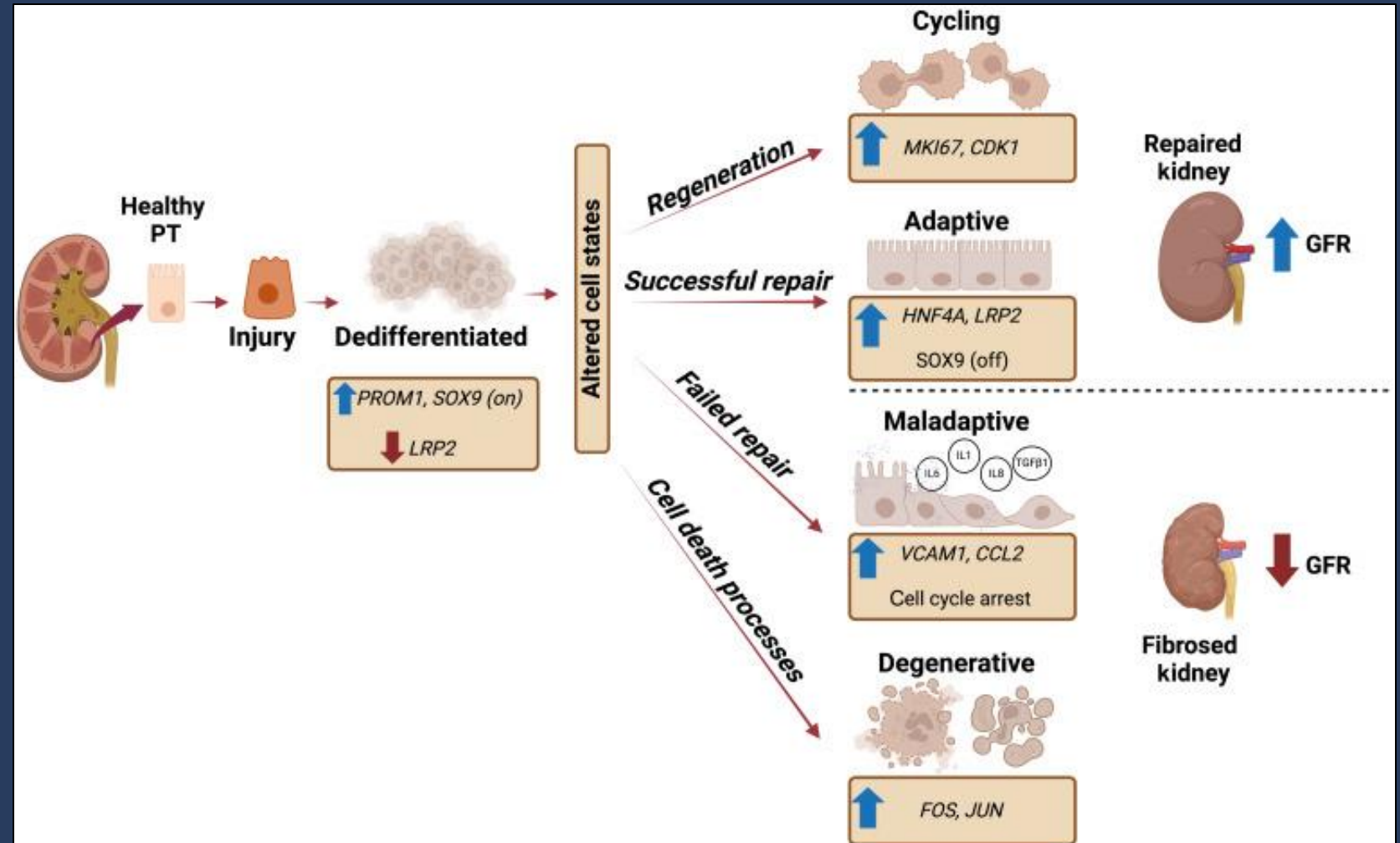
- CODEX spatial proteomic data (~200 samples)
- 3D Tissue Cytometry (~150 samples)
- Visium Spatial Transcriptomics (~200 samples)
- Now providing FFPE CODEX and Xenium ST

Major outputs

- Co-corresponding authors of KPMP Atlas v1 (Nature, 2023, cited 670 times in 2 years)
- KPMP / HuBMAP Atlas of the Renal Papilla
- KPMP / HuBMAP Epigenome Atlas
- KPMP Atlas V2 (under review)

Cell state trajectories in the Kidney Precision Medicine Project Atlas

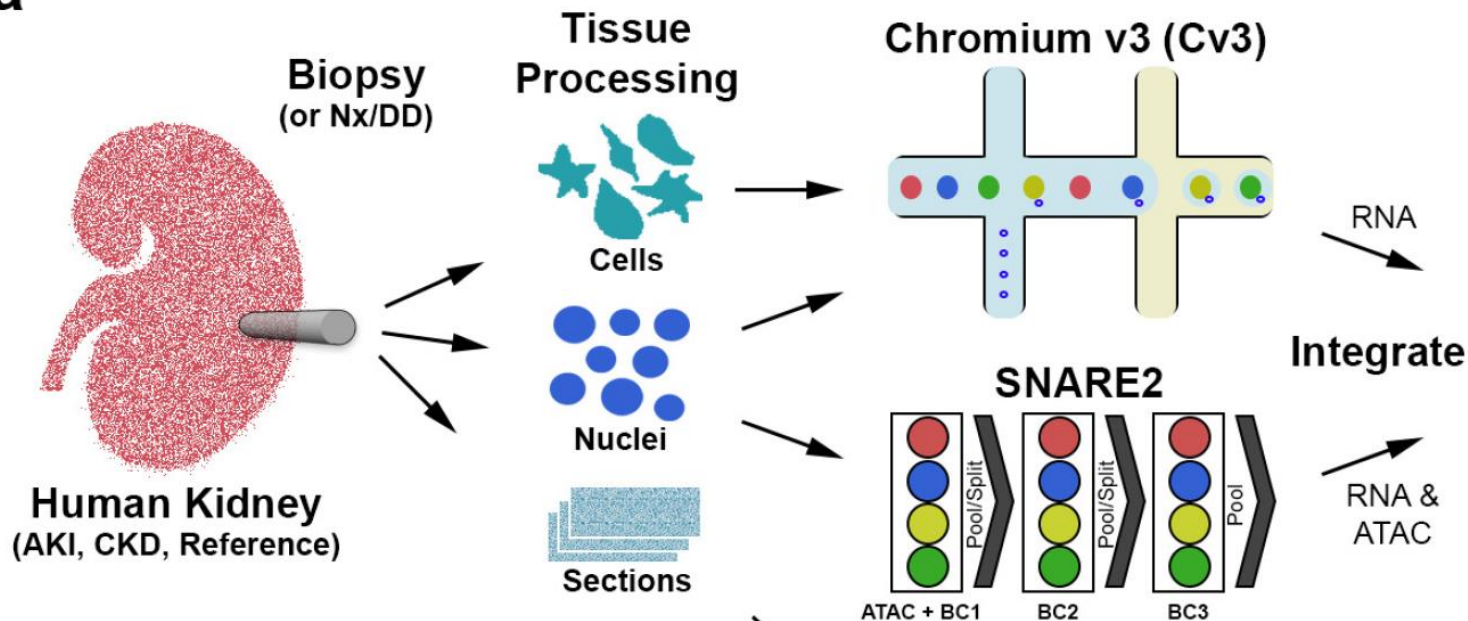
- Epithelial Cell
- Endothelial Cell



Kidney Cell Atlas Overview: Technologies

100 cell types defined, 6 cell states, 122 samples

a

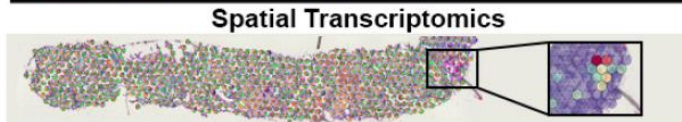
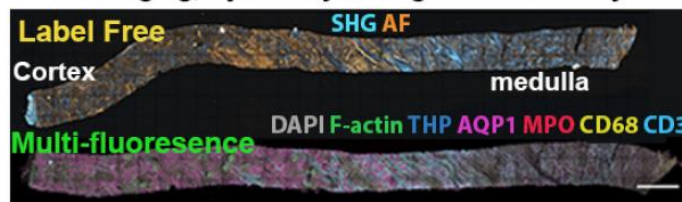


b

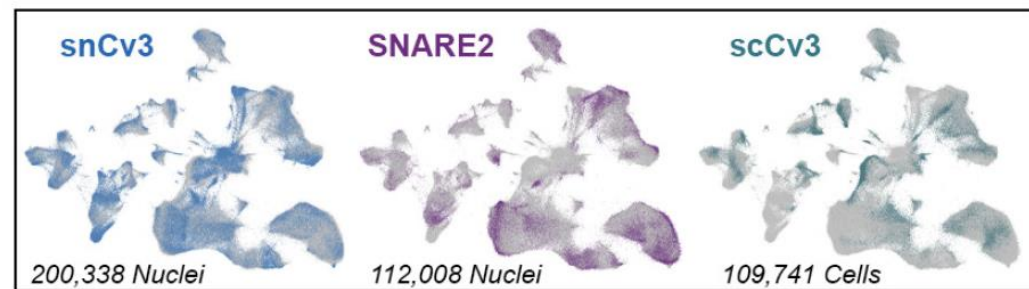
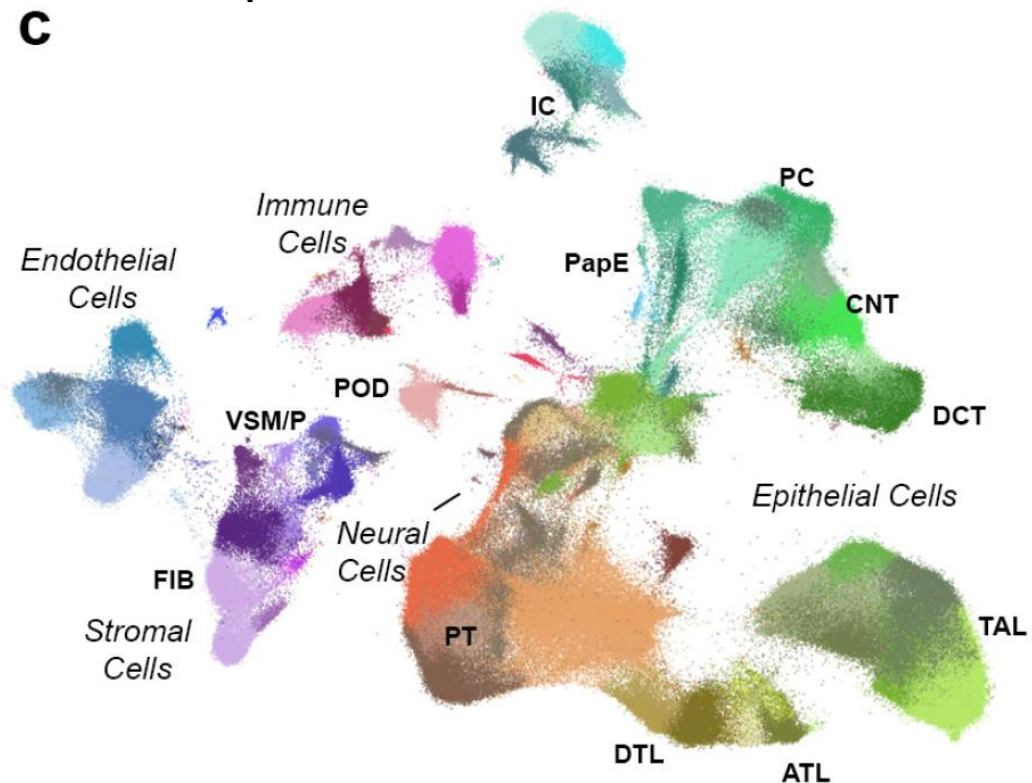
Assay	Sex		Condition		
	♂	♀	CKD	AKI	Ref
10X snRNA-Seq	19	17	13	10	13
10X scRNA-Seq	21	24	15	12	18
SNARE-Seq2	3	4	-	-	7
3D Imaging	6	9	9	6	-
Slide-seq2	5	1	-	-	6
Visium	12	10	10	6	6

Spatial Technologies

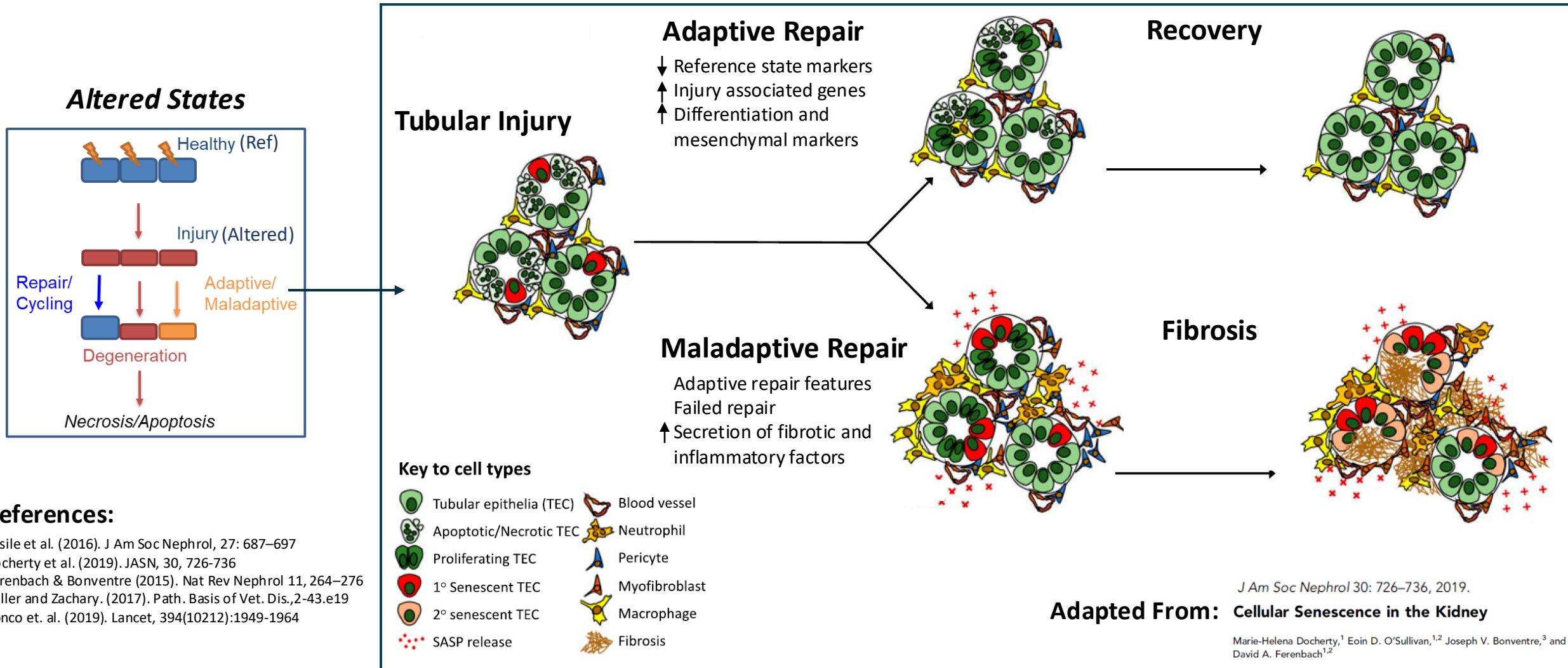
3D Imaging, Cytometry & Neighborhood Analysis



c



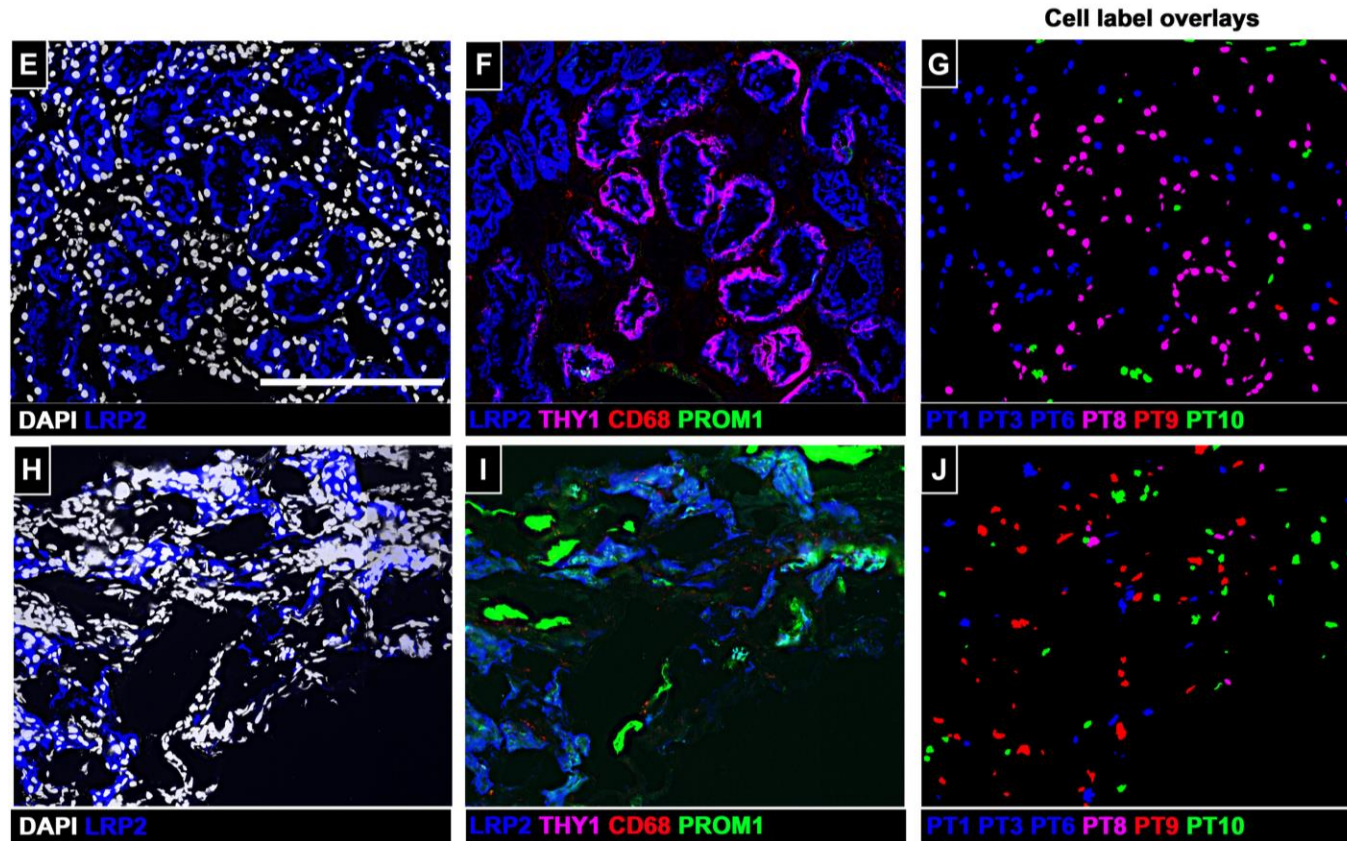
Defining Altered States in Injury



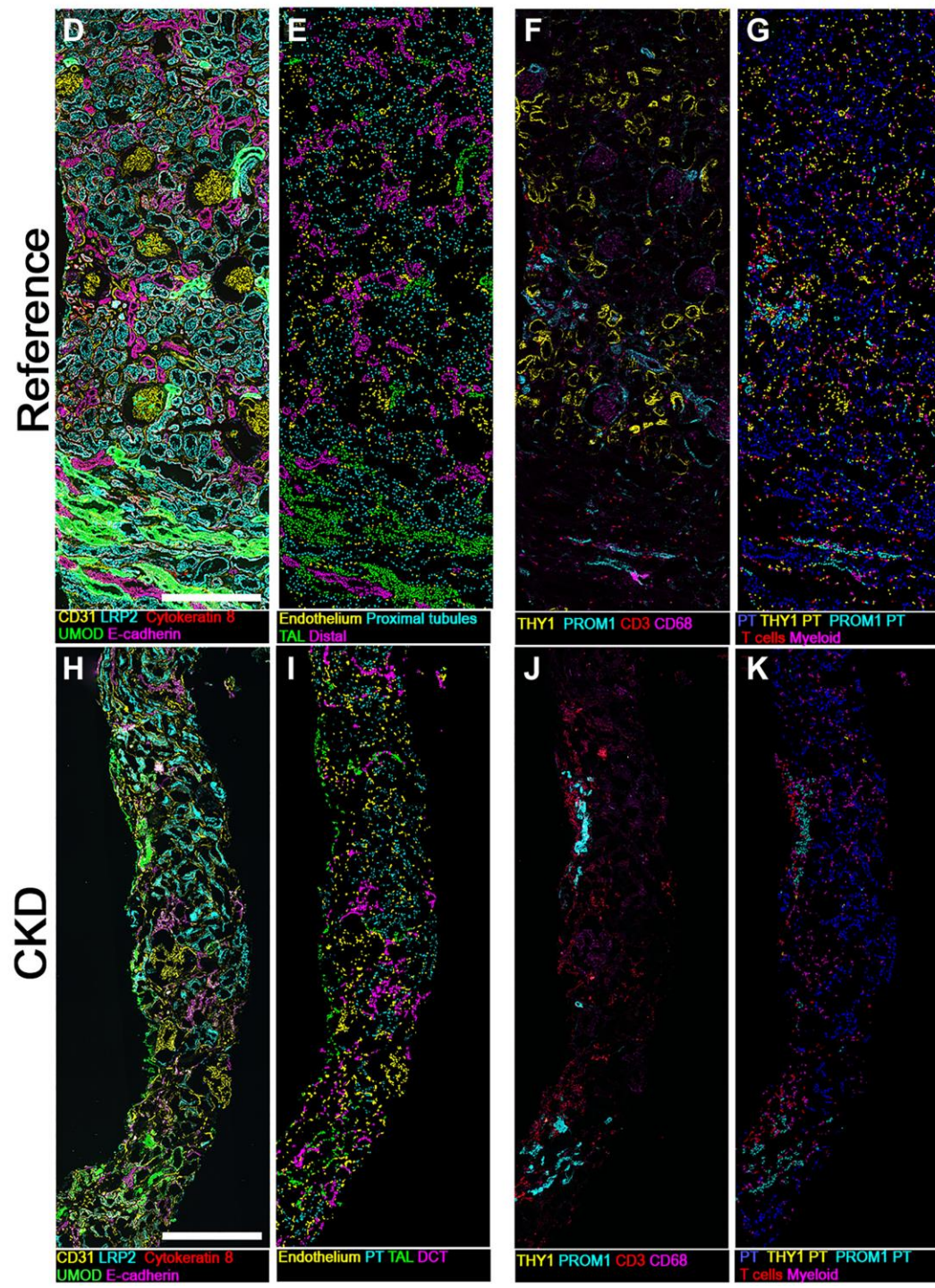
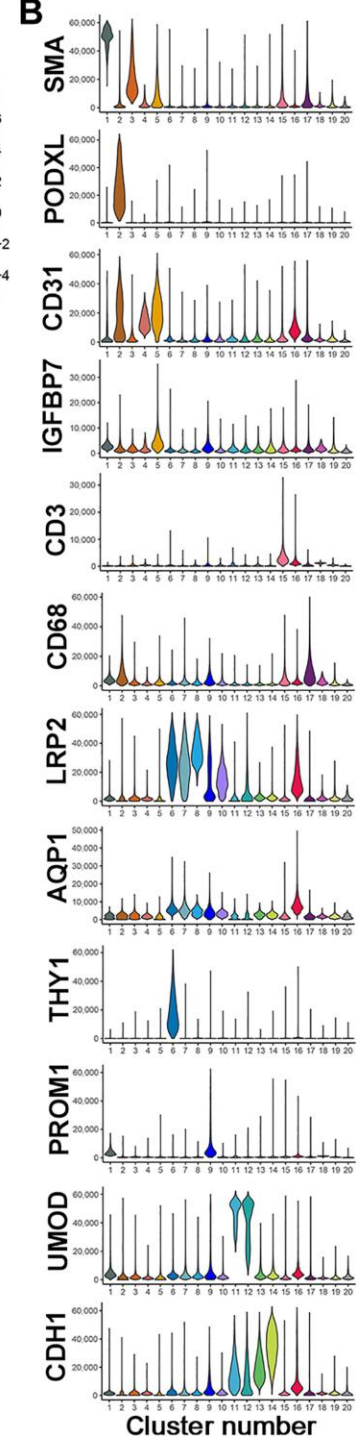
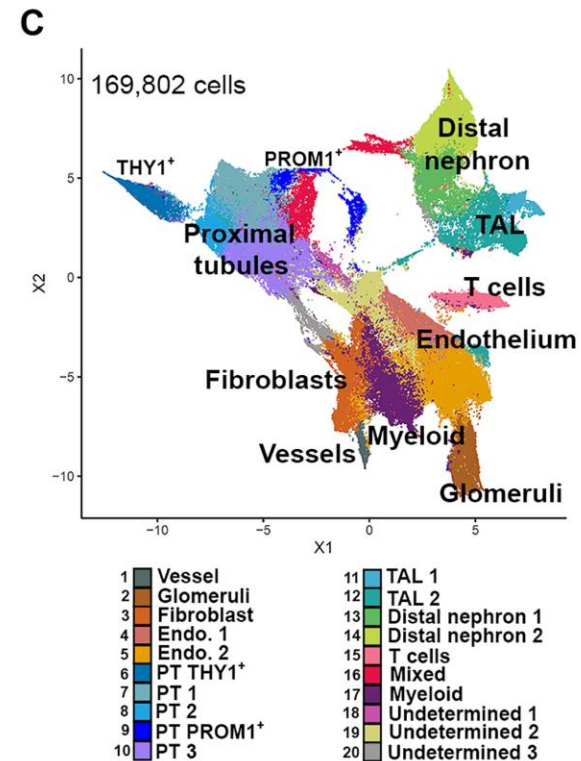
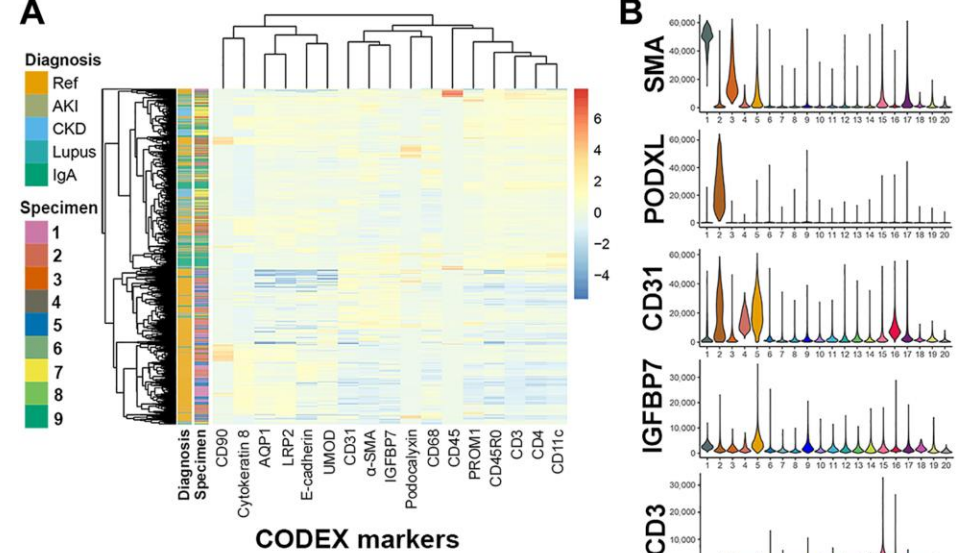
Protein signature drives identification of an under-represented regenerative PT THY1+ cluster



Mahla
Asghari

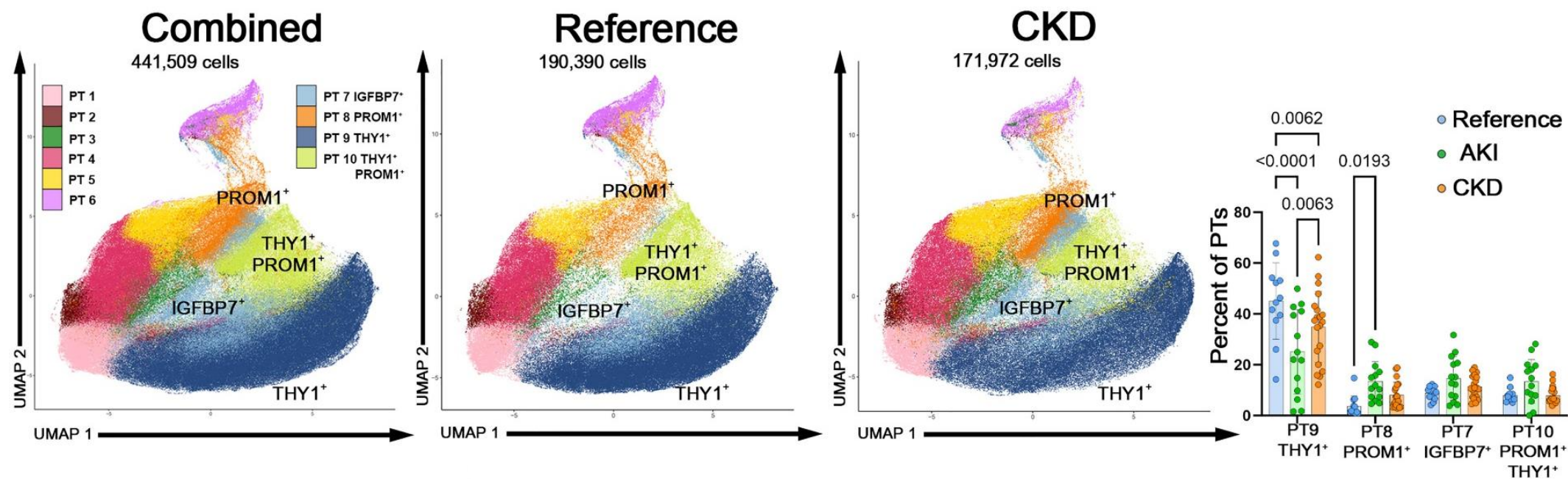


- Examined 2 cell “states”:
- PROM1+ undifferentiated or adaptive PTs
- THY1+ PTs with regenerative potential

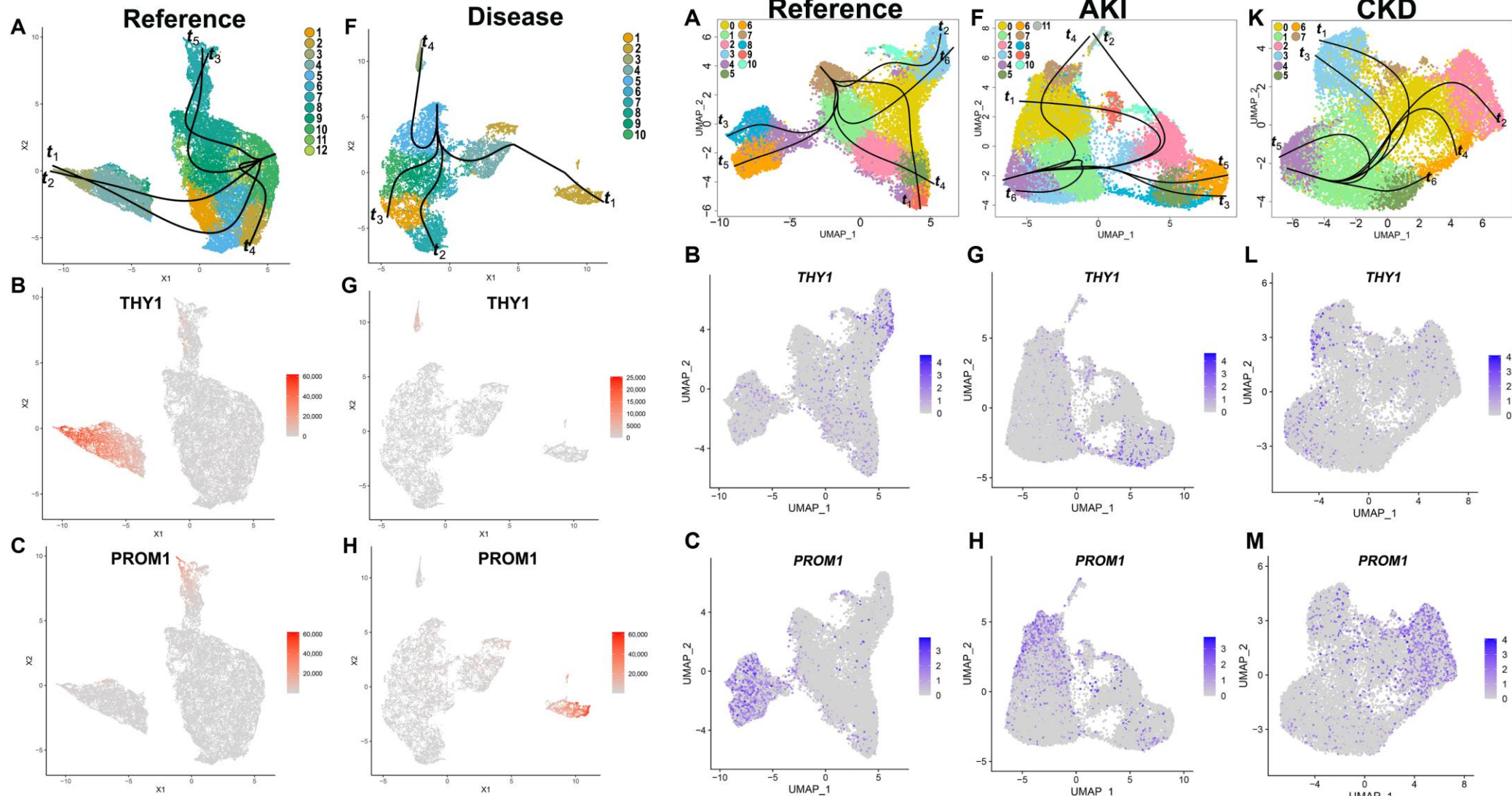


- Integrated a vast amount of CODEX data into a common analytical space

Changes of THY1+ and PROM1+ PTs using spatial protein imaging



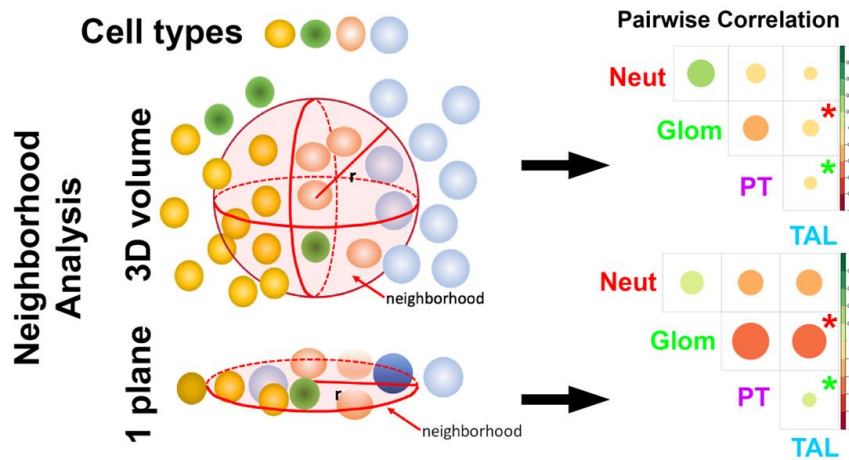
Parallel Protein and RNA trajectories



Protein-based Trajectories

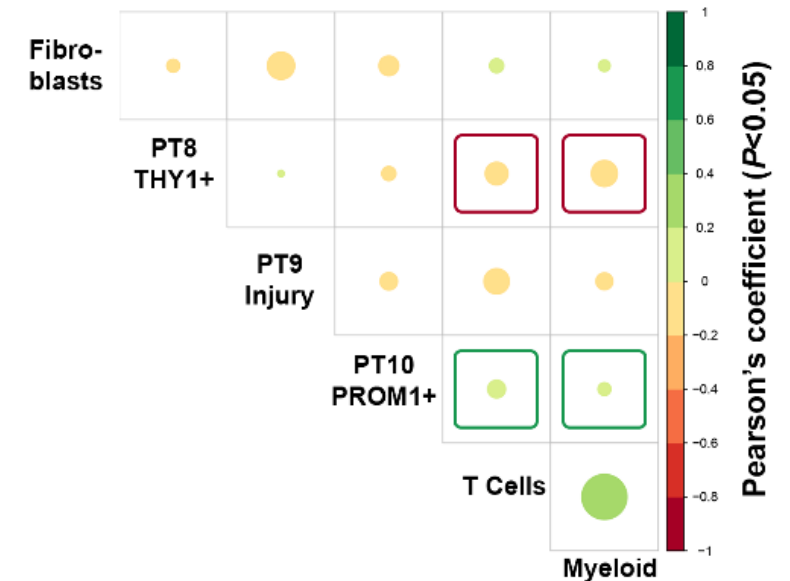
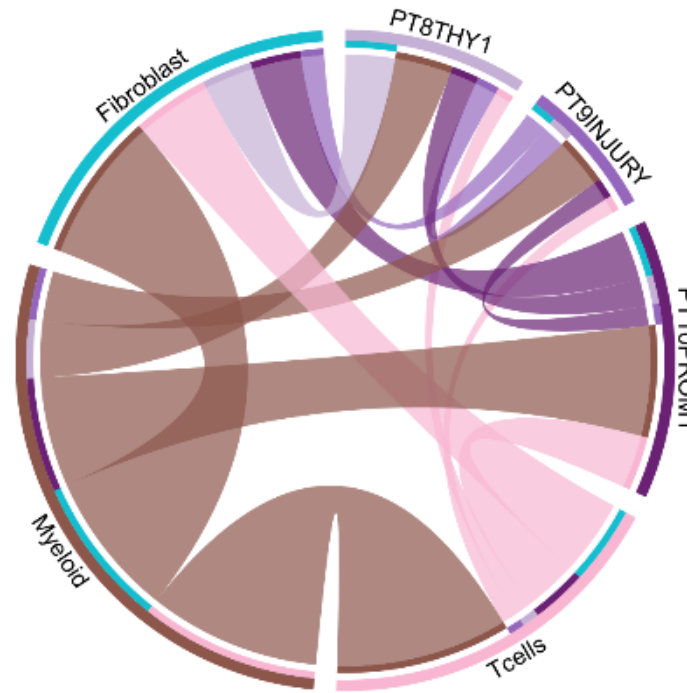
snRNA-seq Trajectories

Neighborhood analysis highlights important insights into pathogenesis

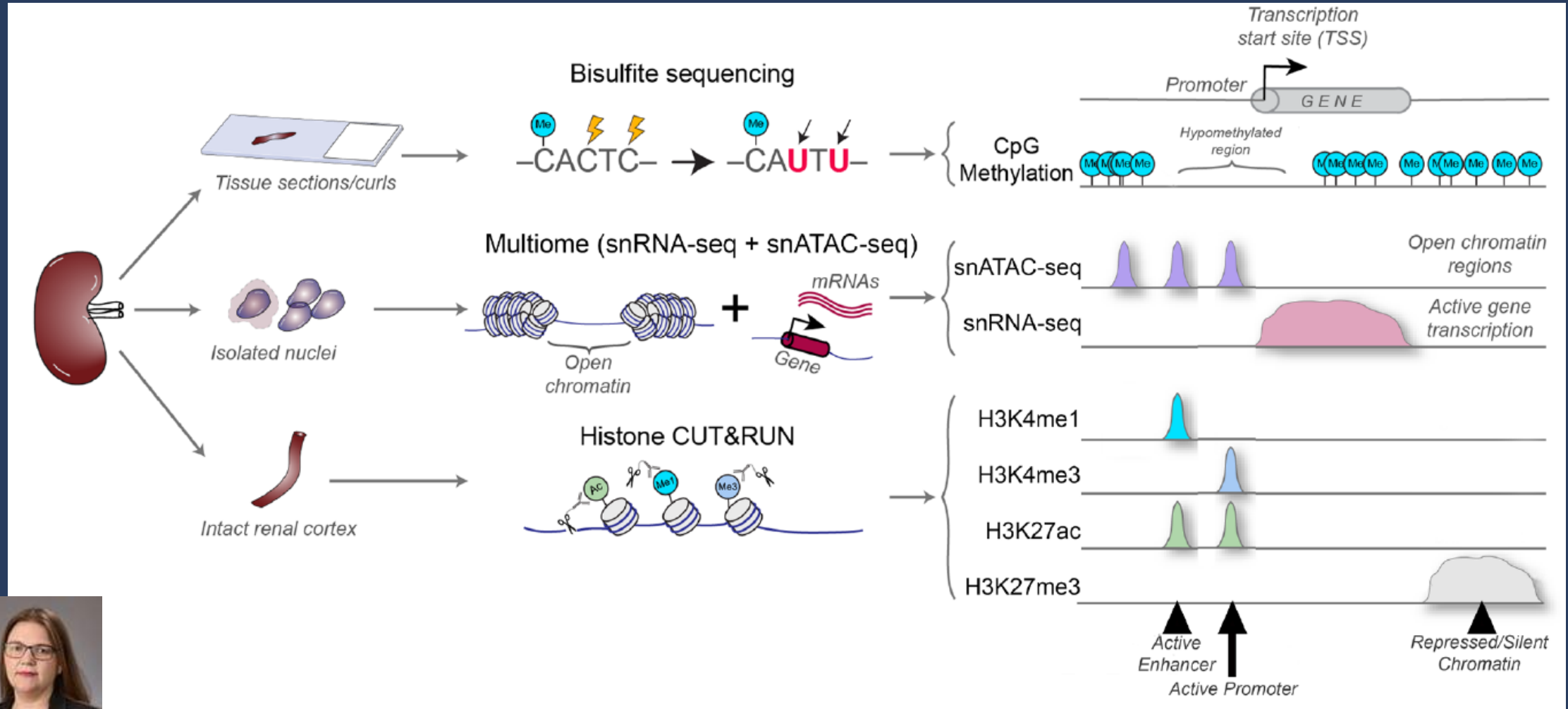


THY1+ PTs negatively associated with immune cells

PROM1+ PTs positively associated with immune cells

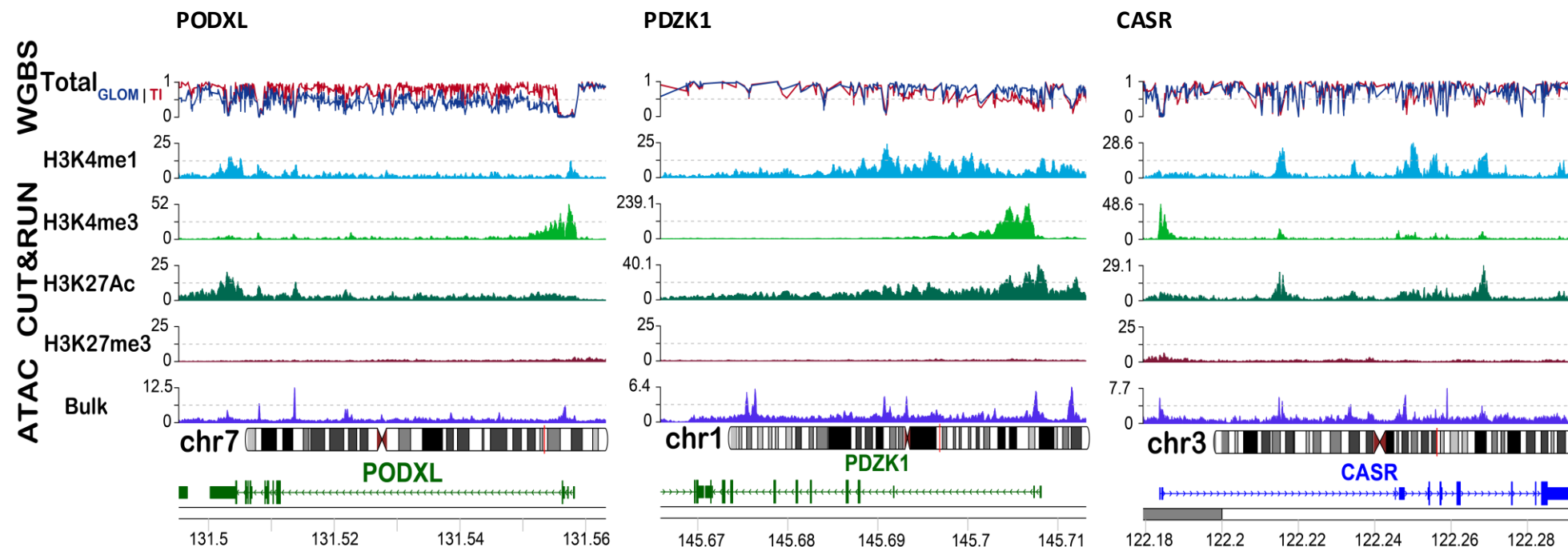


Regulation: Cell State Trajectories and TF Networks



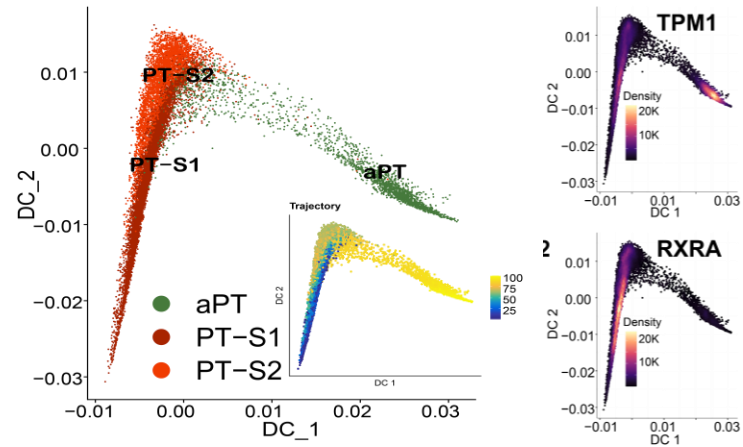
Debora Gisch

Alignment across Genome

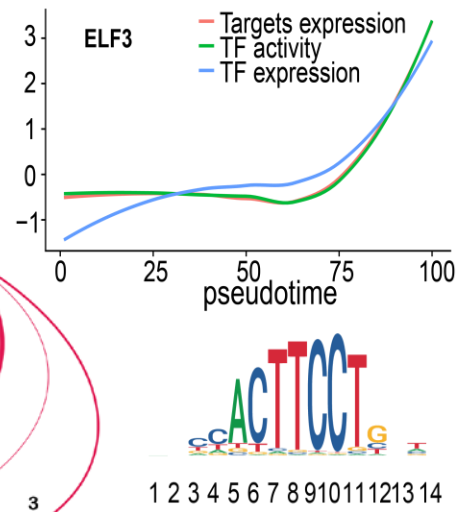
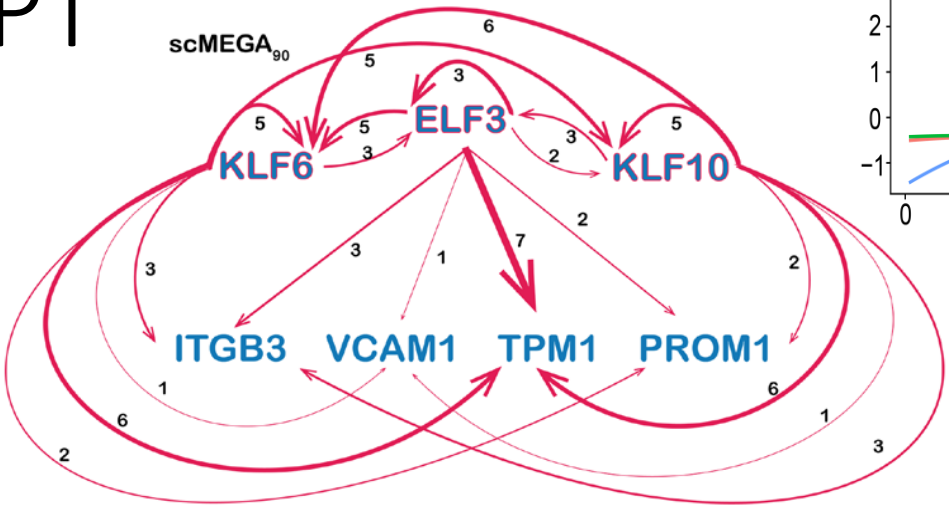


- H3K27ac – activating mark
- H3K27me3 – silencing mark
- H3K4me1 - enhancer
- H3K4me3 - promoter
- Bulk ATACseq (Encode) + Multiome
- WGBS of Glom and TI – 30 samples
- Kidney chromatin landscape browser: <https://www.kpmp.org/doi-collection/10-48698-hhe6-yv15>

ELF3 TF network of the aPT

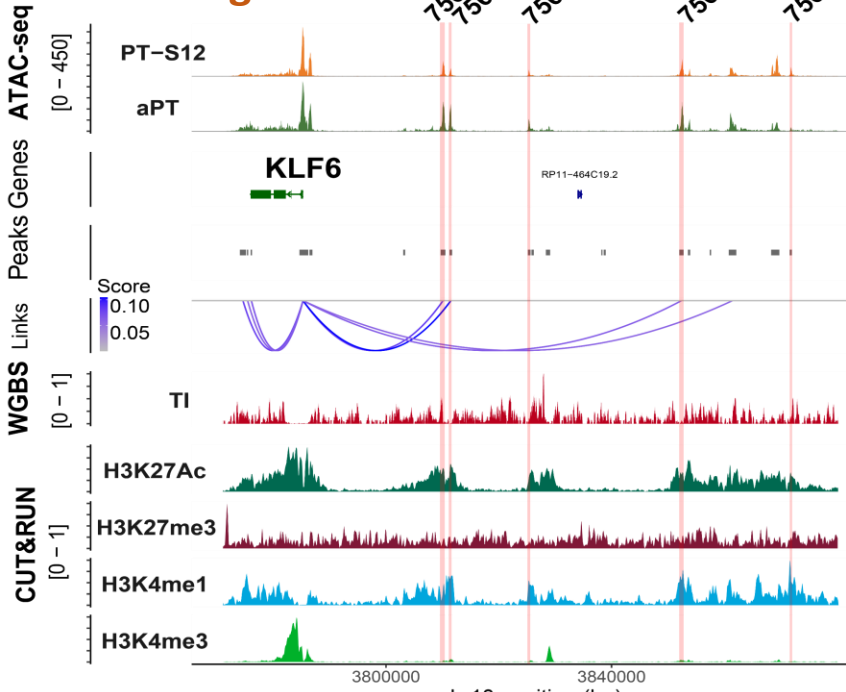


Establish Cell State Trajectories and TF networks

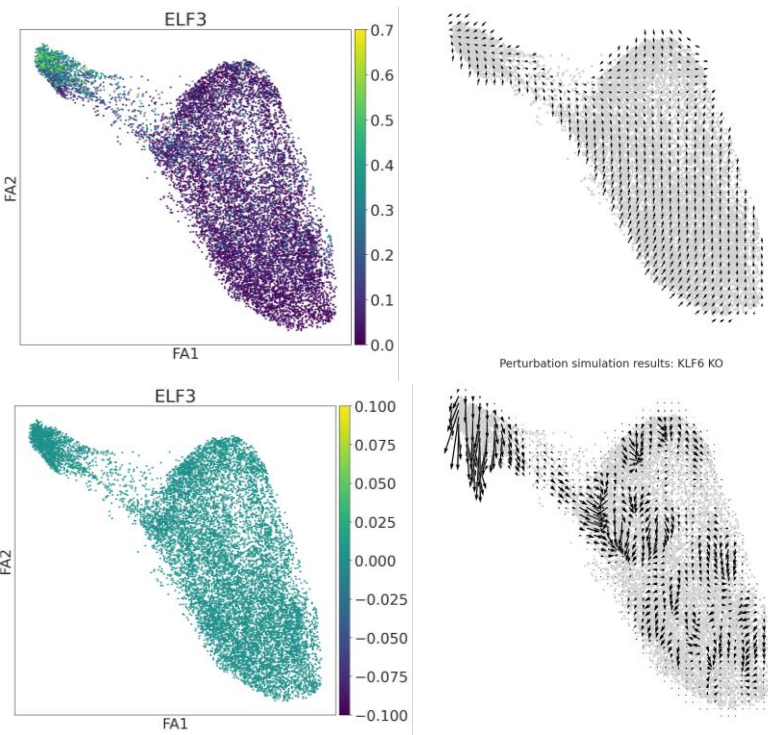


Gene: KLF6 TF: ELF3

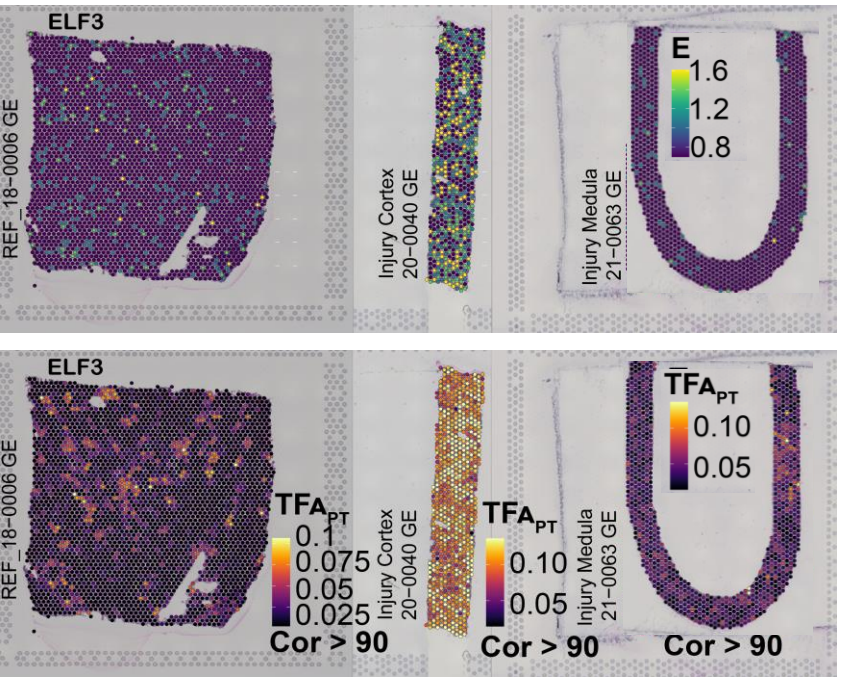
Genome Alignment



In silico Knockdown

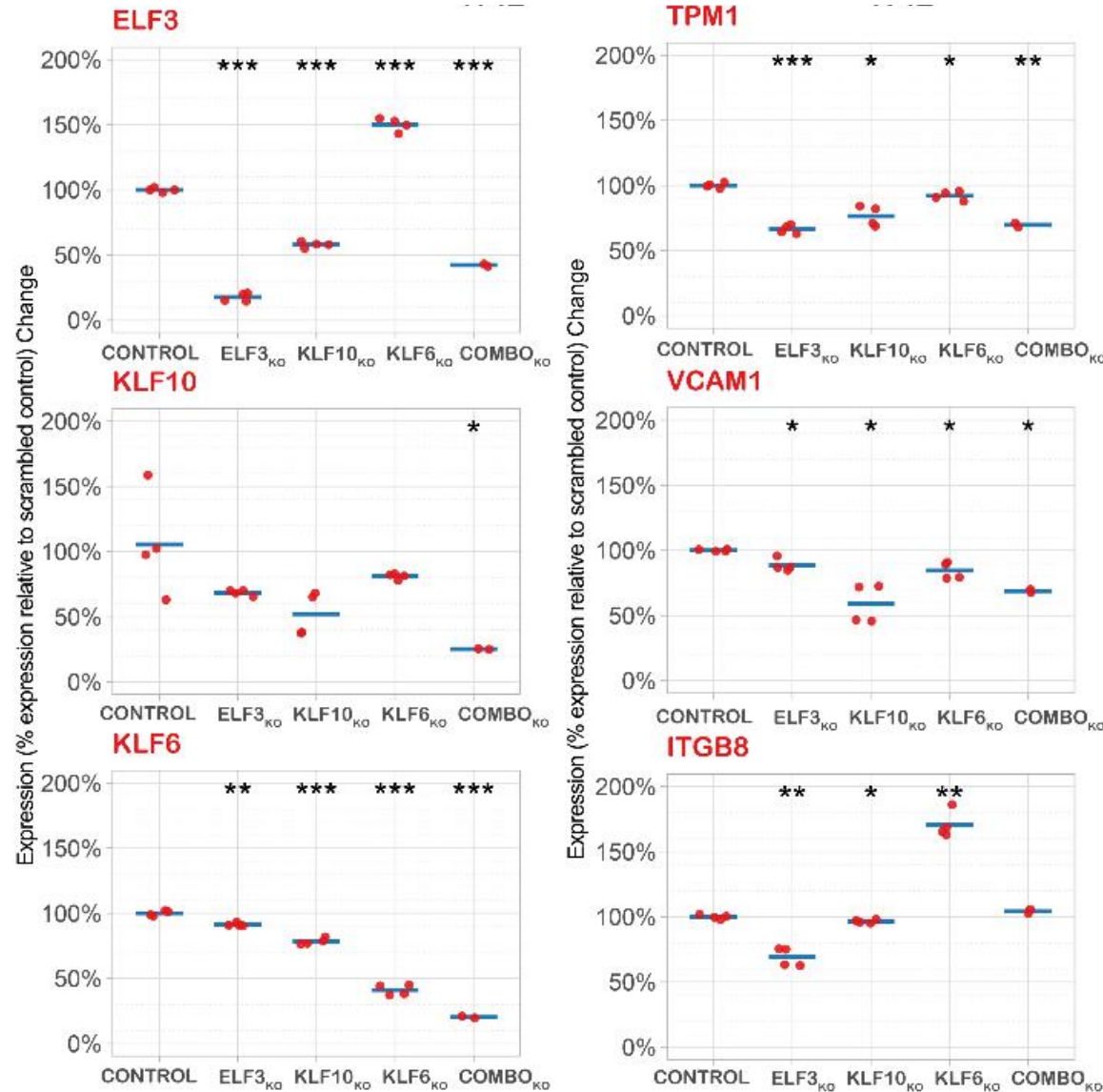


Transcription Factor Activity Mapping



75684 - chr10-3809682-3810536
75685 - chr10-3811268-3811731 75686 - chr10-3825164-3825639
75691 - chr10-3852025-3852799 75696 - chr10-3871597-3871997

siRNA Validation



- Human NHPTK cells
- hTERT immortalized
- Express Megalin
- KO of TF network
- Assessment of target gene expression



MOLECULAR SIGNATURE



CLINICAL & STRUCTURAL



Signature I



Signature II



Cellular and Spatial Drivers of Unresolved Injury and Functional Decline in the Human Kidney

"Atlas 2.0" Publication



Sanjay Jain MD PhD

Unique aspects - HKA expansion: V1 -> V2

Atlas v1



Health
+
Injury/Disease

93 patients
215 samples
~0.3M cells

**40 KPMP
Biopsies**

Atlas v2



Health
+
Injury/Disease
+
Cross species
+
Age
+
Clin_path
+
Noninvasive markers –
progression
+
Mechanisms

353 patients
709 samples
~1.7M cells

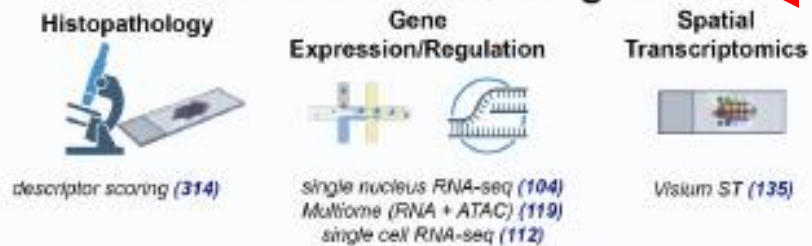
**235 KPMP
Biopsies**

Condition Summary Counts	Total
Acute Kidney Injury	63
Chronic Kidney Disease	150
Diabetes Mellitus - Resilient	12
Healthy Reference	92
Reference	36
Grand Total	353

Sampling Diversity

# participants	Healthy Reference Tissue (HRT)	Acute Kidney Injury (AKI)	Chronic Kidney Disease (CKD)	Other Ref
Sex (M/F/Unk):	43/49	43/20	82/65/3	25/22/1
Race (W/B/O):	80/5/7	36/22/5	61/42/27	33/3/12
Age Range:	10-99	10-79	20-89	20-89
Total:	92	63	150	48

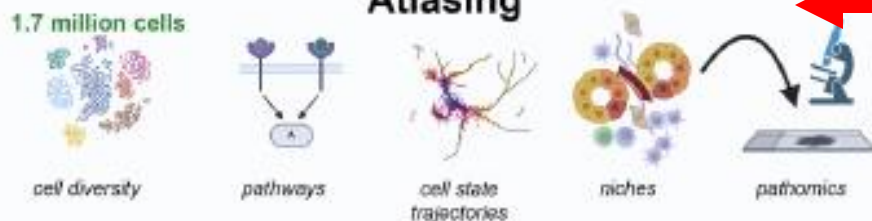
Core Atlas Technologies



Validation Technologies



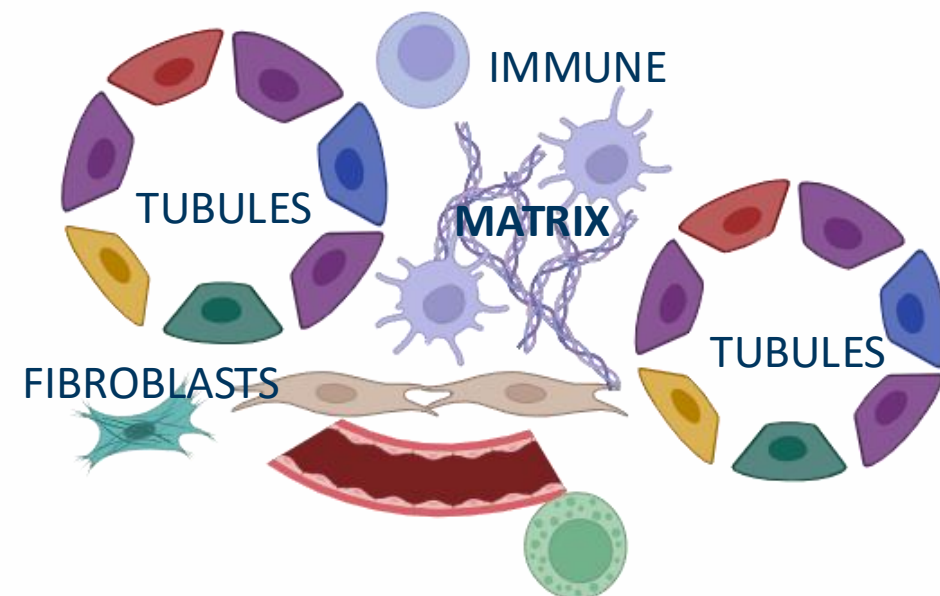
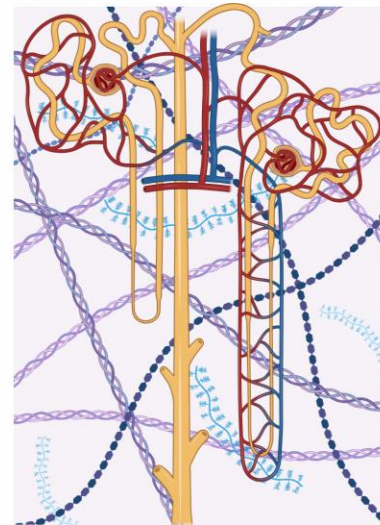
Atlasing



Mechanisms & Clinical Insights



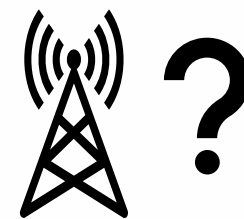
OVERVIEW / APPROACH of HKAv2



TUBULES – Healthy or Injured (**Kidney Function**)

FIBROBLASTS – Supporting infrastructure, adjust depending on tubular state

IMMUNE – Defend or Attack depending on tubular state



HKA v2

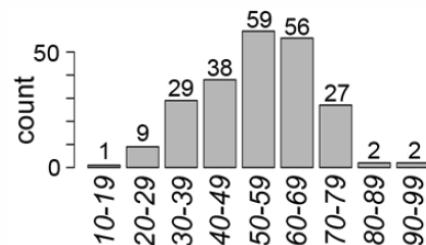
Single Nucleus
RNA/Accessible Chromatin
Sequencing

$n = 233$

Patient

Patient Age

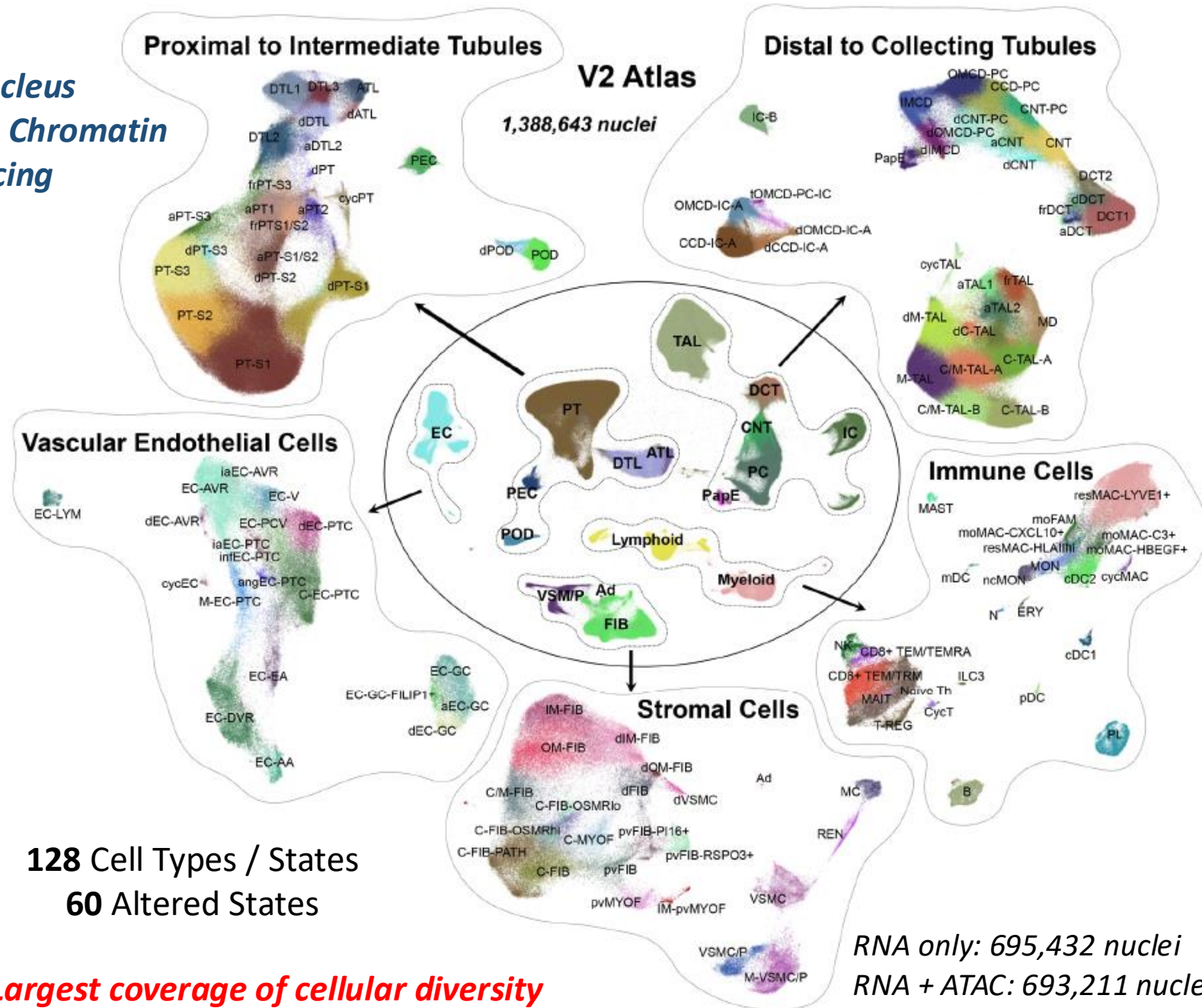
Condition



● HRT ● AKI ● CKD

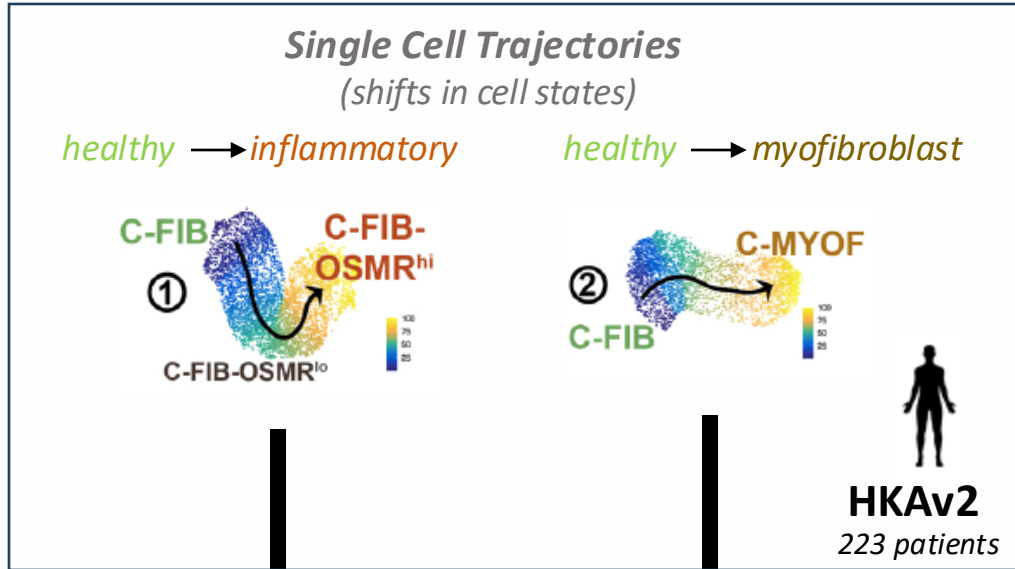
128 Cell Types / States
60 Altered States

Largest coverage of cellular diversity

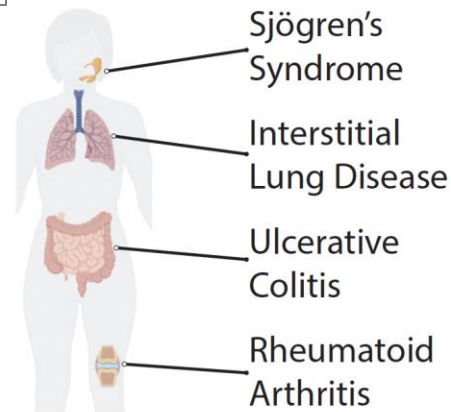
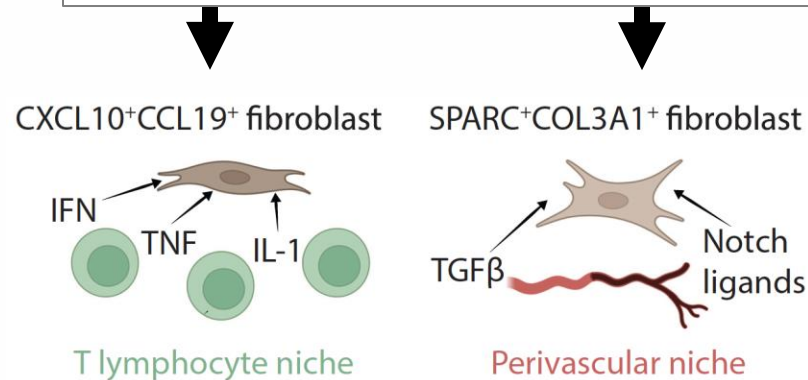


Biology to Disease: Pathogenic Fibroblasts

State Shifts in Kidney Injury/Disease

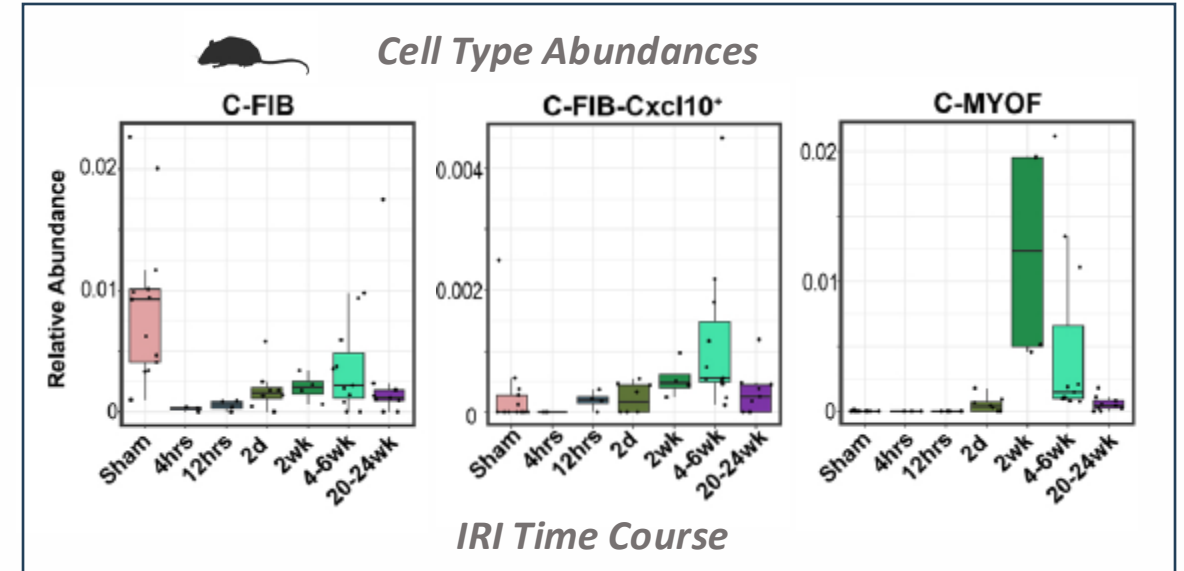


Conserved Across Diseases

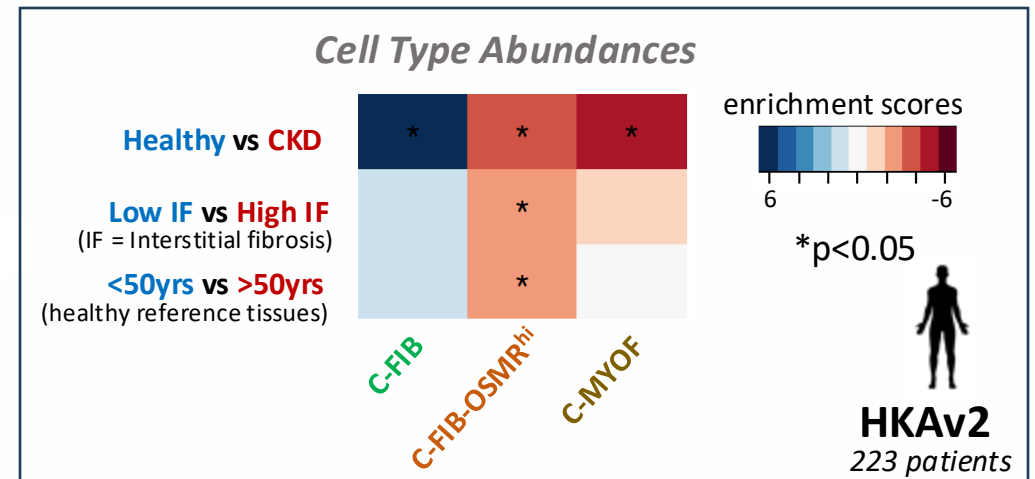


Korsunsky et al., 2022

Conserved in Mouse Models

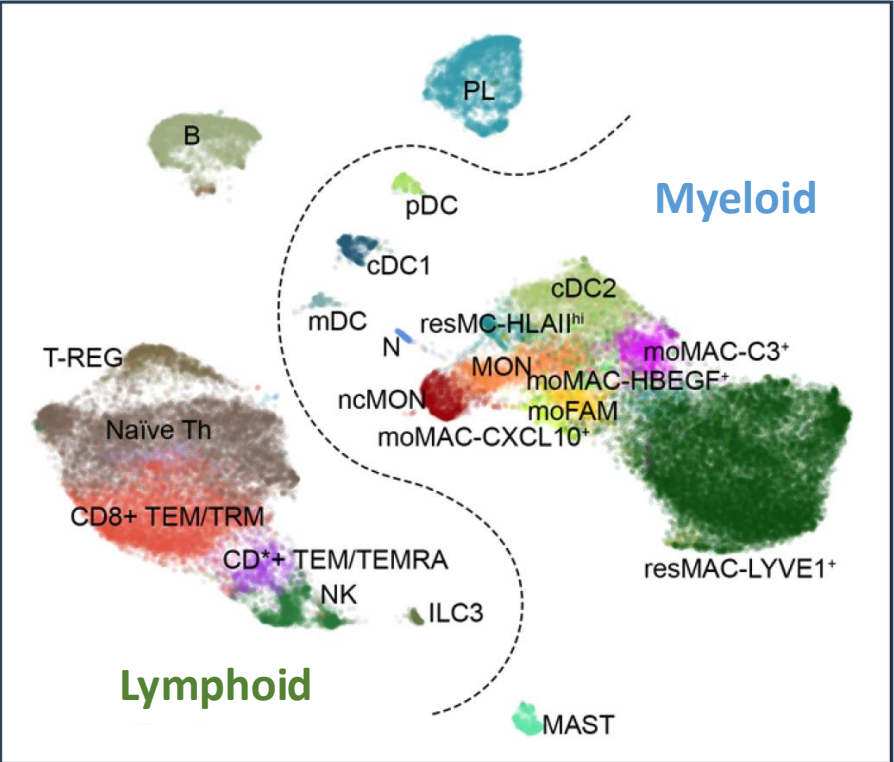


Enriched in Chronic Disease & Aging



Immune Cells in Injury and Disease

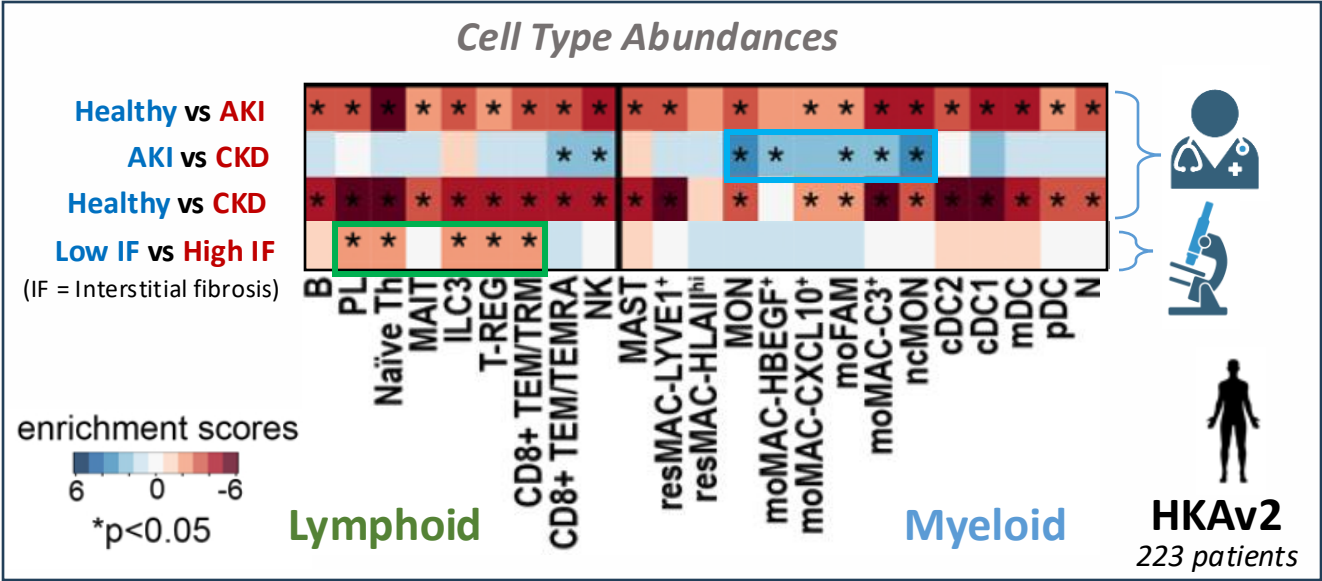
Immune Cell Diversity



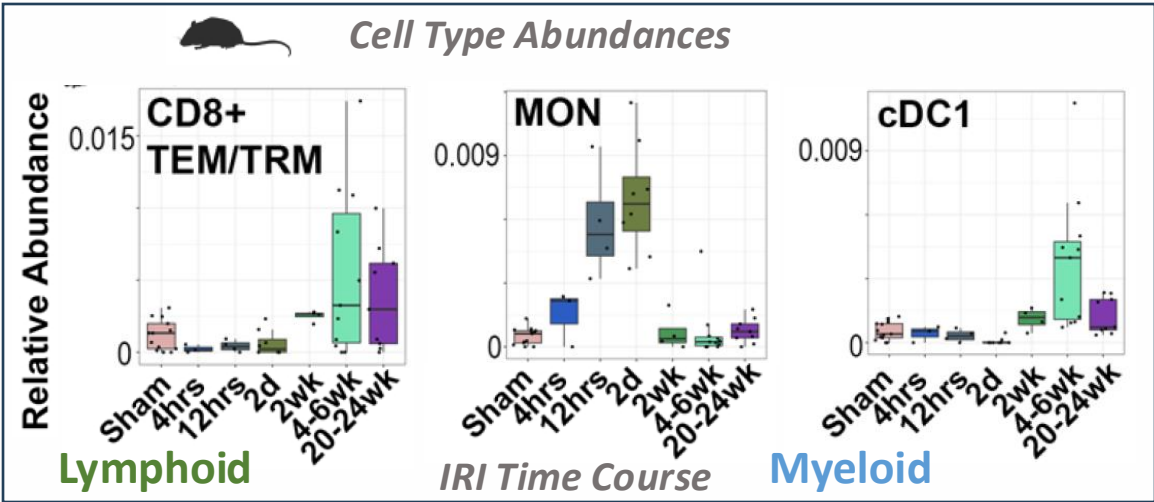
Single cell / Single nucleus Integration: [Rajasree Menon](#)

Lymphoid cells enrich with high fibrosis
Myeloid cells enrich with acute injury

Enriched in Acute and Chronic Disease

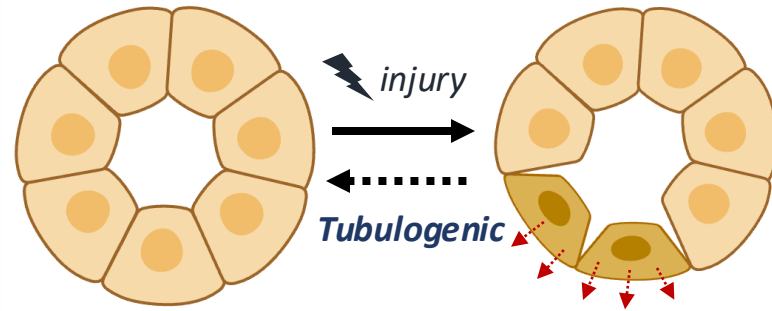


Conserved in Mouse Models

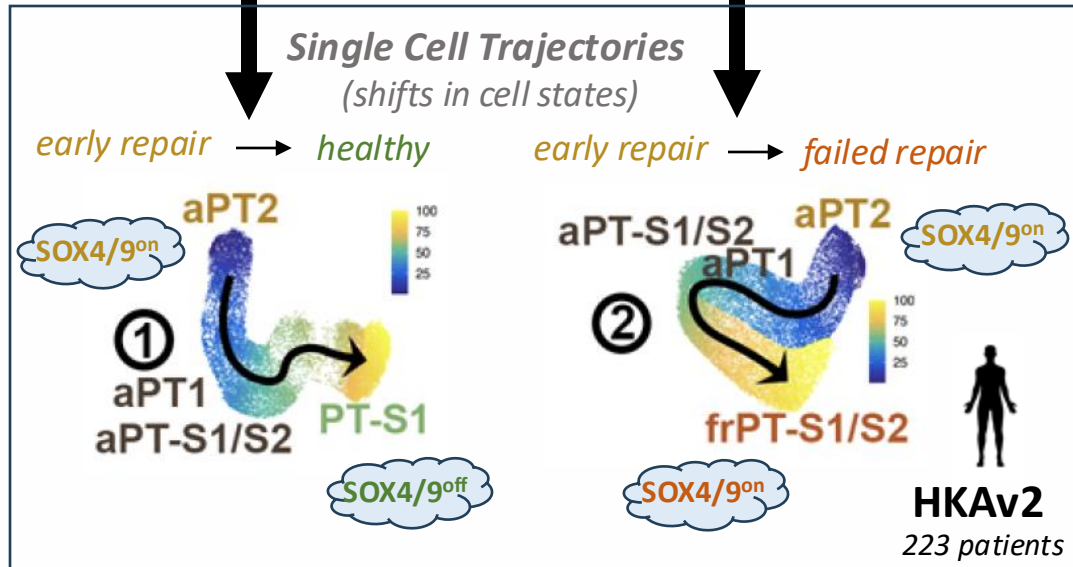


Unresolved Epithelial Repair

Enriched in Acute and Chronic Disease

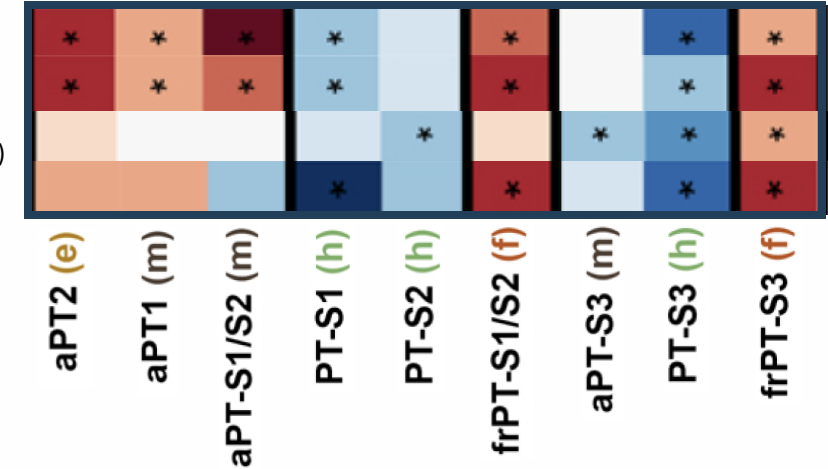


State Shifts in Kidney Injury/Disease

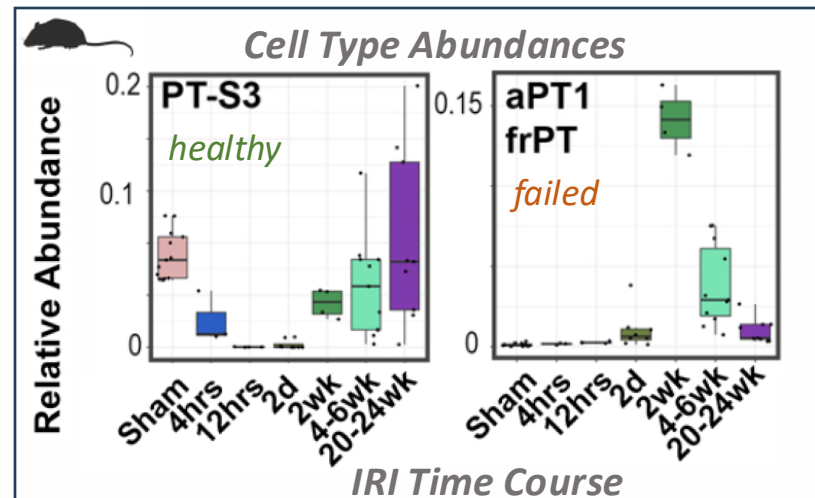


Cell Type Abundances

Healthy vs AKI
Healthy vs CKD
<50yrs vs >50yrs
(healthy reference tissues)
Low IF vs High IF
(IF = Interstitial fibrosis)

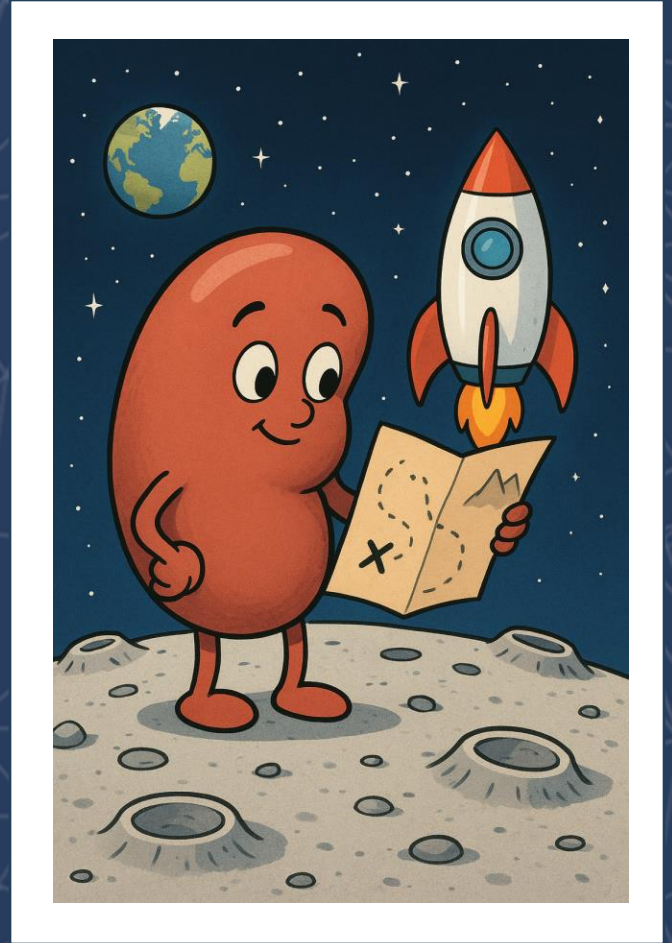


Conserved in Mouse Models



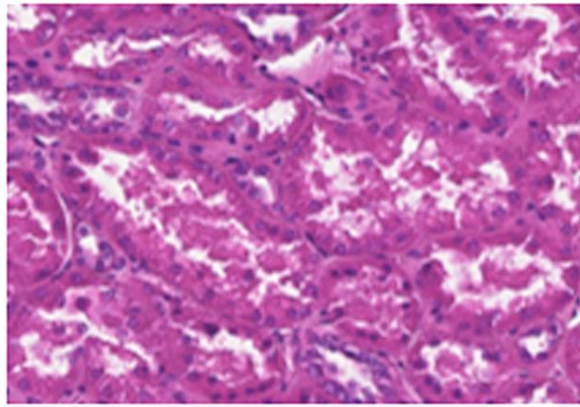
Unresolved (failed) epithelial repair is enriched with aging, CKD and fibrosis

Kidney Atlas v2 in Space

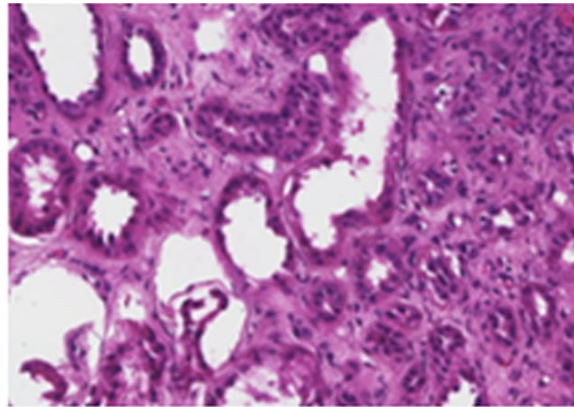


Spatial Localization in the Kidney Atlas v2

Histologic Annotation of Kidney Biopsy Samples



Healthy Kidney



Fibrotic Kidney

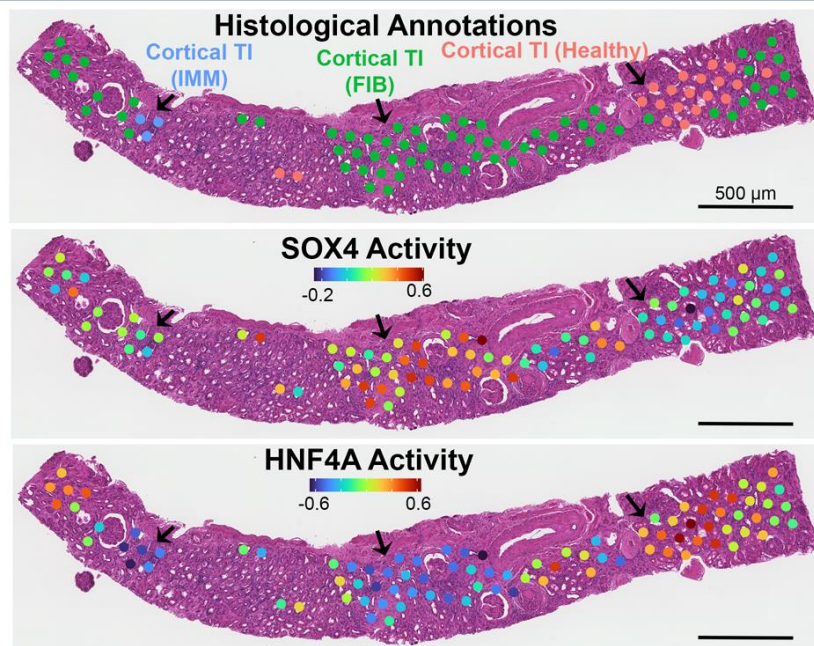
How do the molecular signals we identified organize spatially?

What are the histologic implications for disease progression?

Pathology links
to molecular
signature

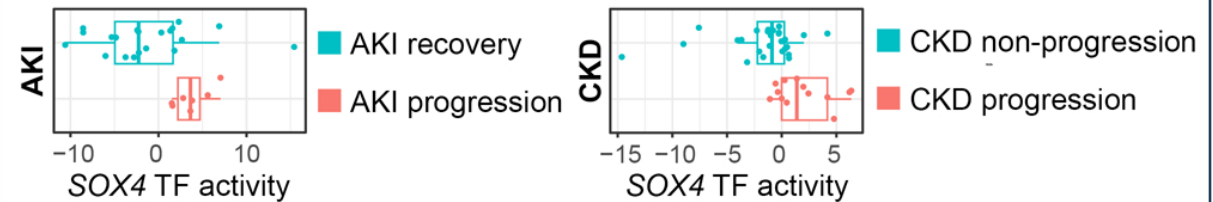


Ricardo Melo Ferreira



**SOX4 Network mapping associated with
AKI and CKD outcome in KPMP heroes**

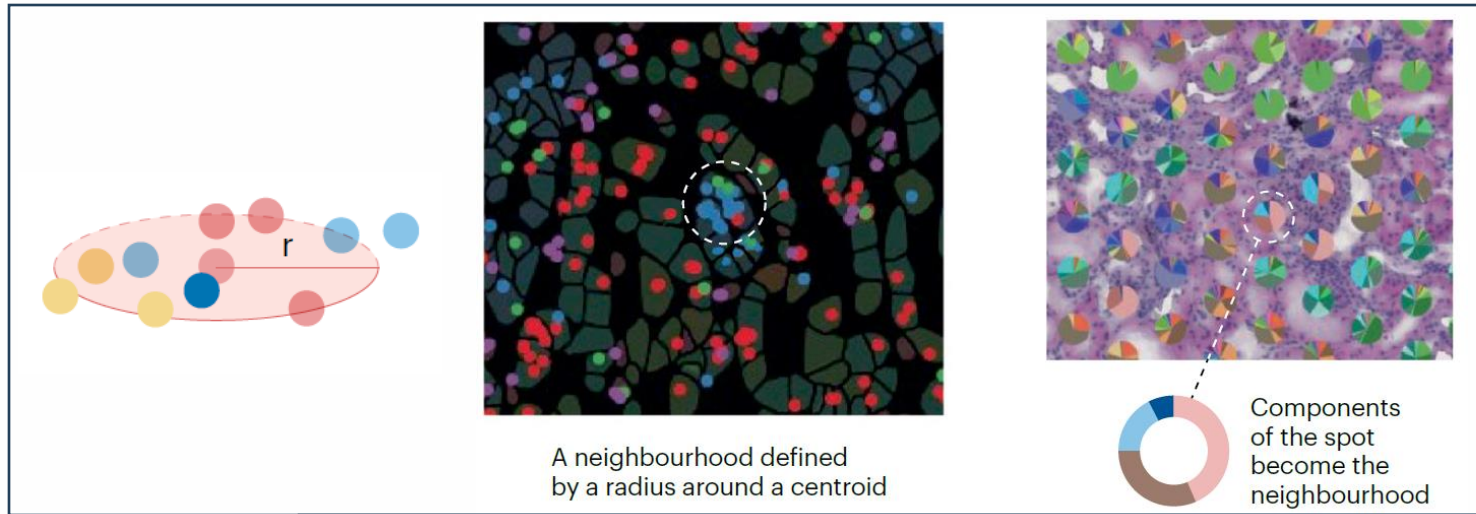
SOX4 TF activity and outcomes



Common feature of AKI and CKD

Neighborhoods

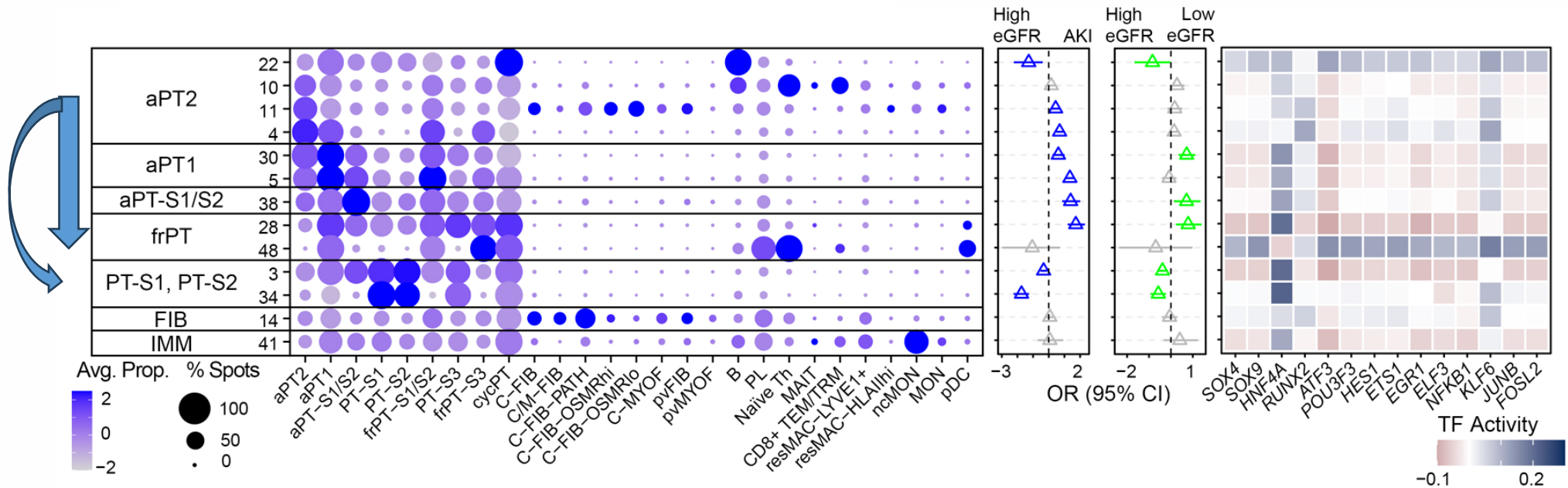
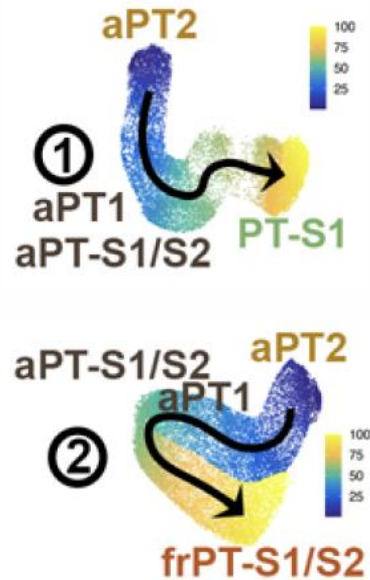
Neighborhoods defined by cells close to each other



Which groups of cells are more common in disease?

What molecular signals correspond to each group of cells?

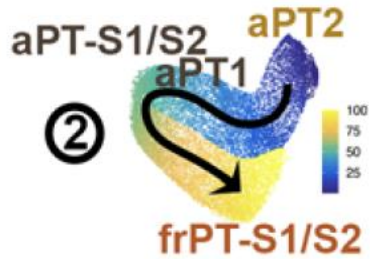
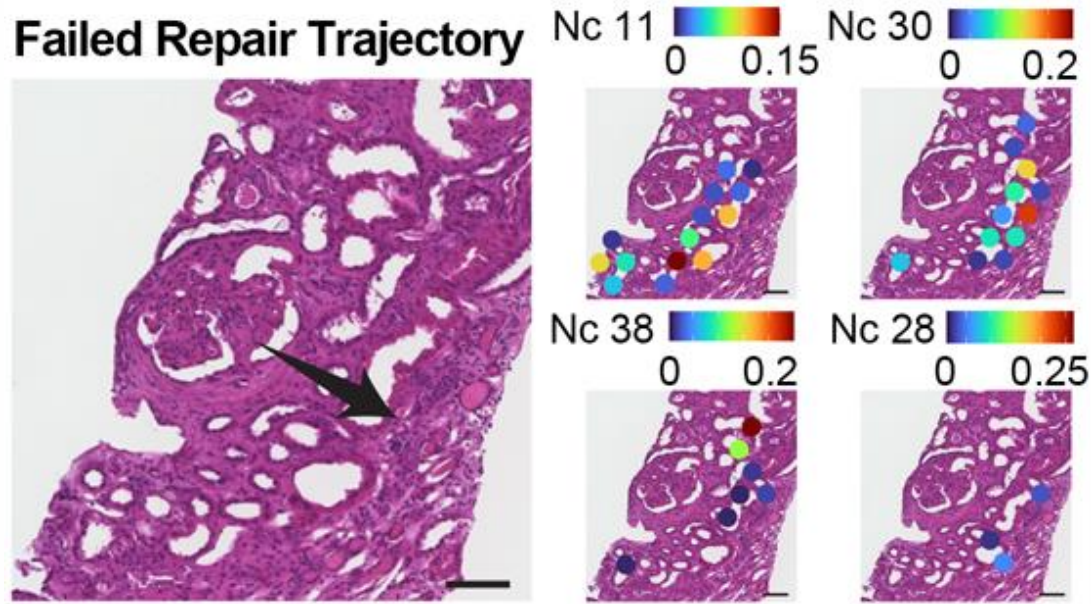
Is there a spatial time course?



Neighborhoods

Temporal neighborhoods follow a spatial trajectory

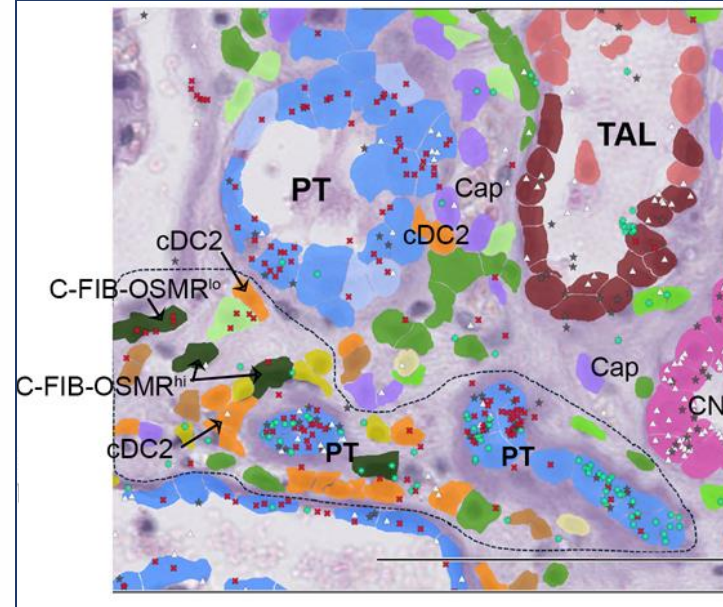
Failed Repair Trajectory



Spatial distribution of cell groups follow cell injury timelines

Ricardo Melo Ferreira

Single Cell Neighborhoods



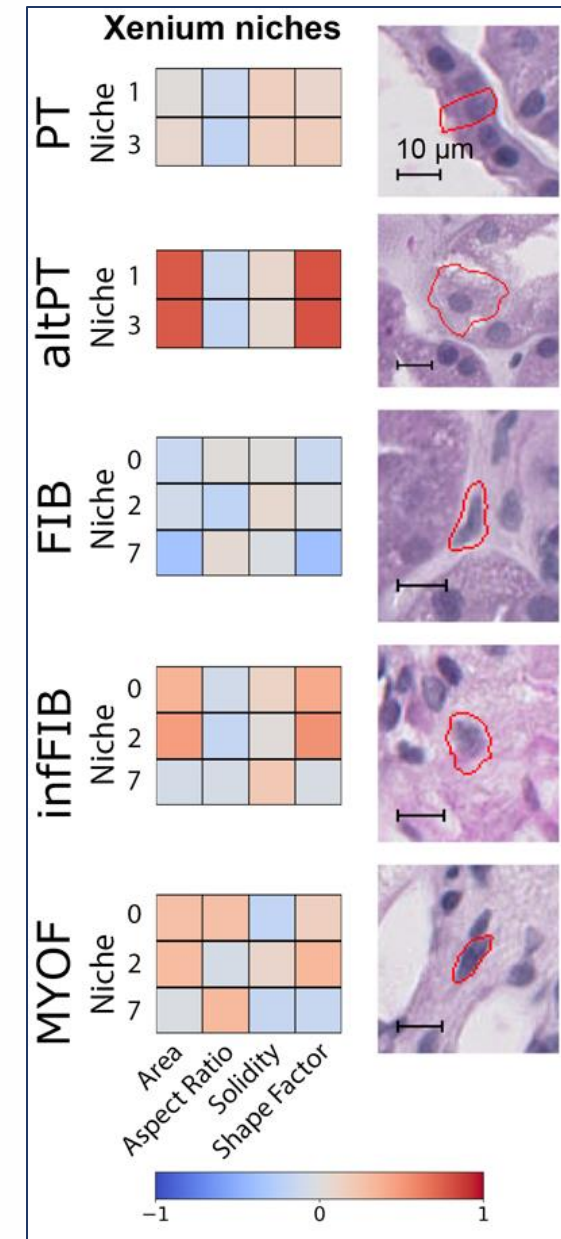
Stephanie Reinert, Asmita Lagwankar

Cells within a neighborhood validated at the single cell level.

Cells expressing SOX4 are seen more frequently in CKD.

Healthy and injured cells look different

AI Cell Features

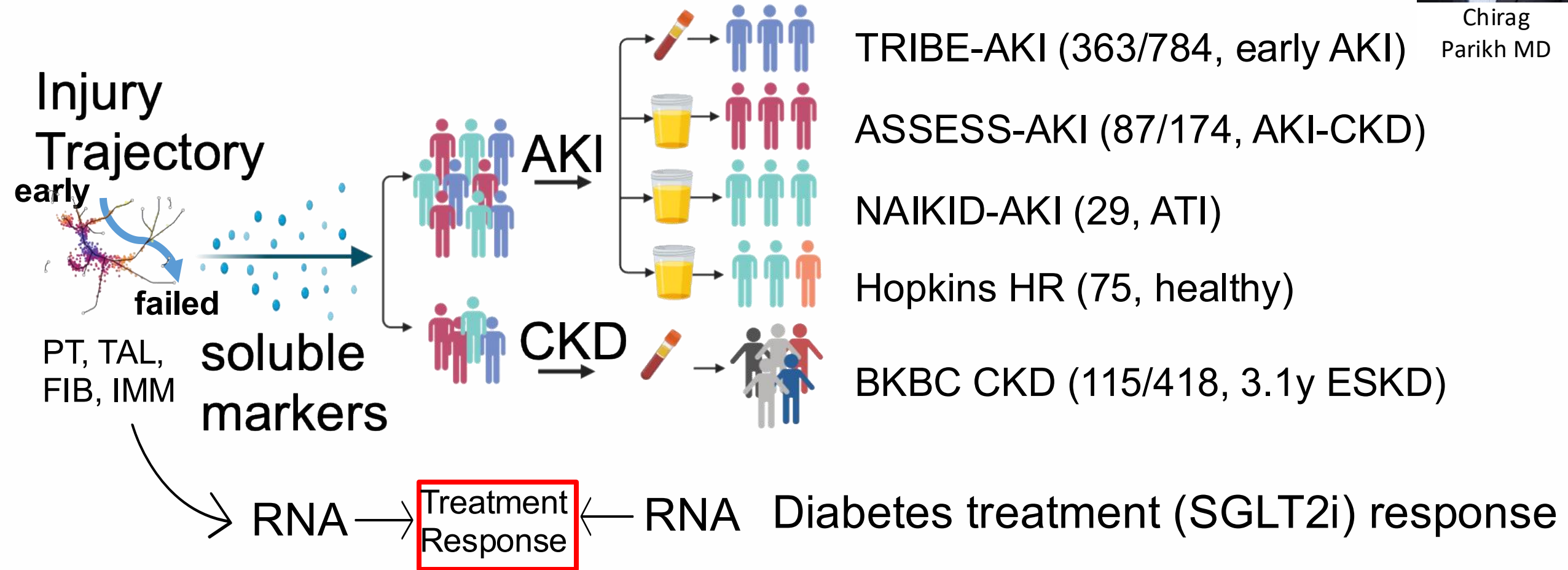


Nick Lucarelli, Pinaki Sarder

Clinical Insights: Early Markers Predict Decline or Recovery after Kidney Injury

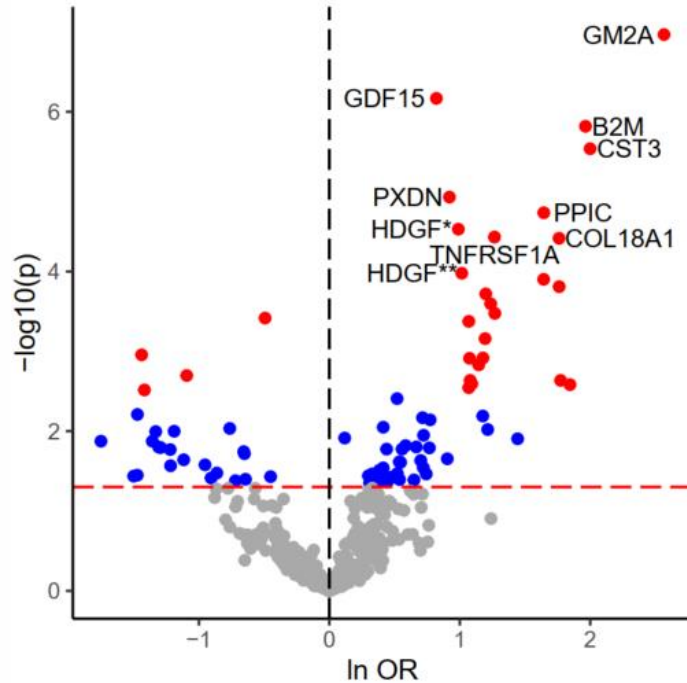


Chirag
Parikh MD



RESULTS [Application to External Datasets]

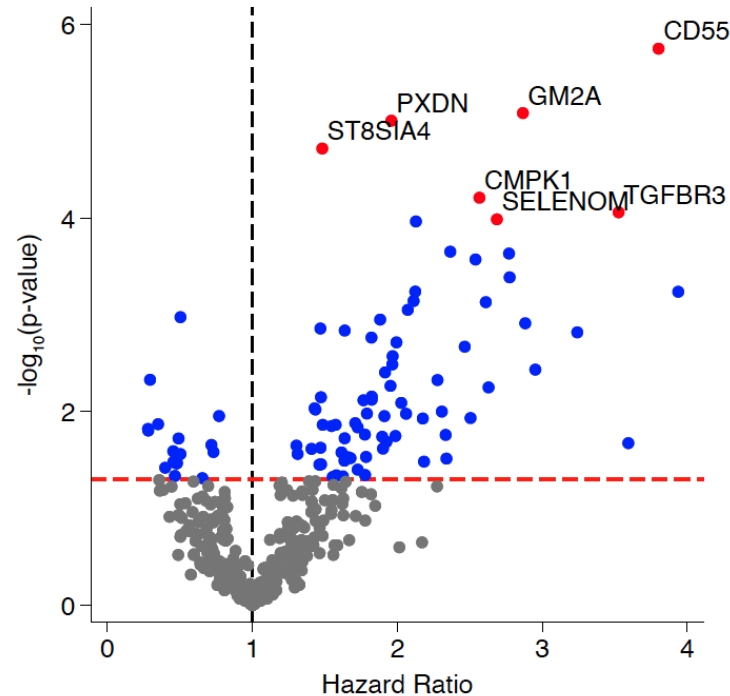
AKI severity



27 proteins were significantly associated with AKI in the TRIBE-AKI Cohort

Heather Philbrook

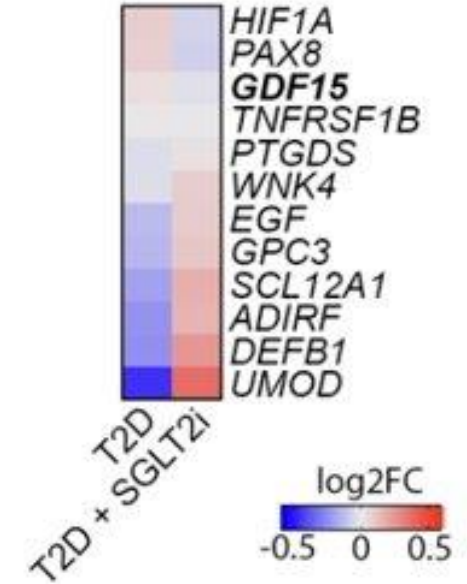
CKD progression



82 significant, 7 pass stringent BH criteria

Insa Schmidt, Sus Waikar

Drug Response (TAL)



Viji Nair, Petter Bjornstard, Mathias Kretzler

- **Early injury secreted proteins associated with decline in kidney function**
- **Several novel proteins modulate fibroimmune environment and matrix**

- **Used cell state markers to identify early treatment response to diabetic medicine**

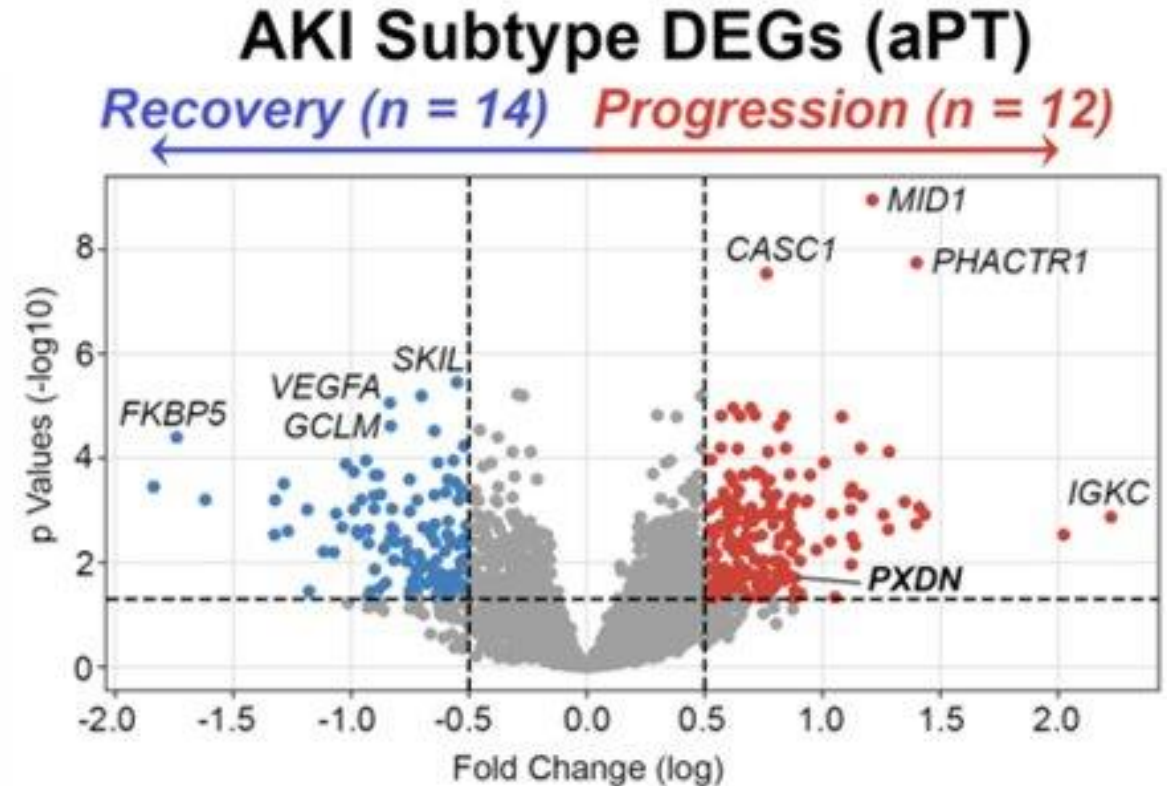
Red dots represent significant proteins (BH)
Blue dots not significant by BH with $p < \text{Bonferroni } (0.05/417)$

Grey dots not significant ($p > 0.05$)

CKD Incidence or Progression - KPMP Cohort (18mo follow up)



- The ASSESS-AKI consortium definition of CKD Incidence or Progression was applied.
 - CKD Incidence defined as at least a 25% reduction in eGFR and a fall below 60 mL/min per 1.7m²
 - CKD progression was defined as at least a 50% reduction in eGFR or a fall below 15 mL/min per 1.73m²
- Follow-up Creatinine (18 months):
 - All outpatient serum creatinine values were considered first, followed by the last available inpatient or unknown source serum creatinine
- Of the 26 participants profiled by snRNA-seq, 14 had AKI recovery and 12 had CKD incidence or progression.

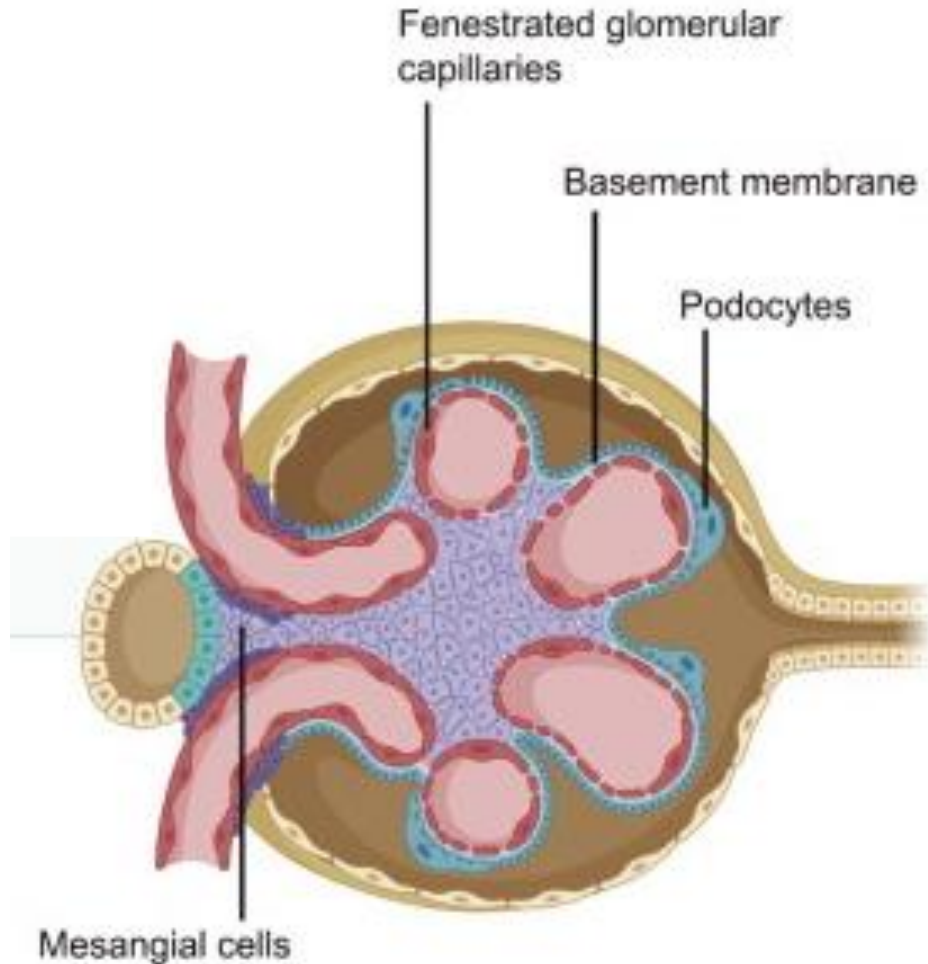


Finding molecules in KPMP biopsies that correlate with decline in kidney function

Endothelial Cells

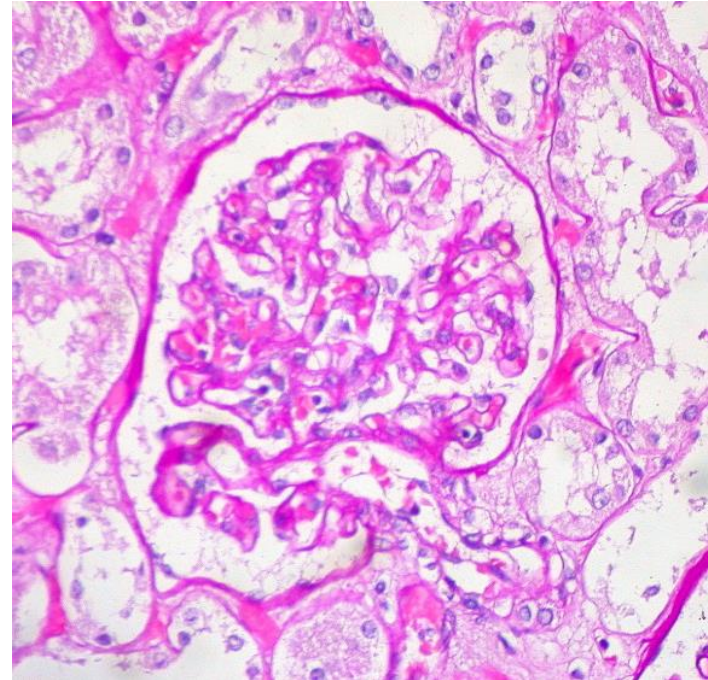


Nodular Mesangial Glomerulosclerosis



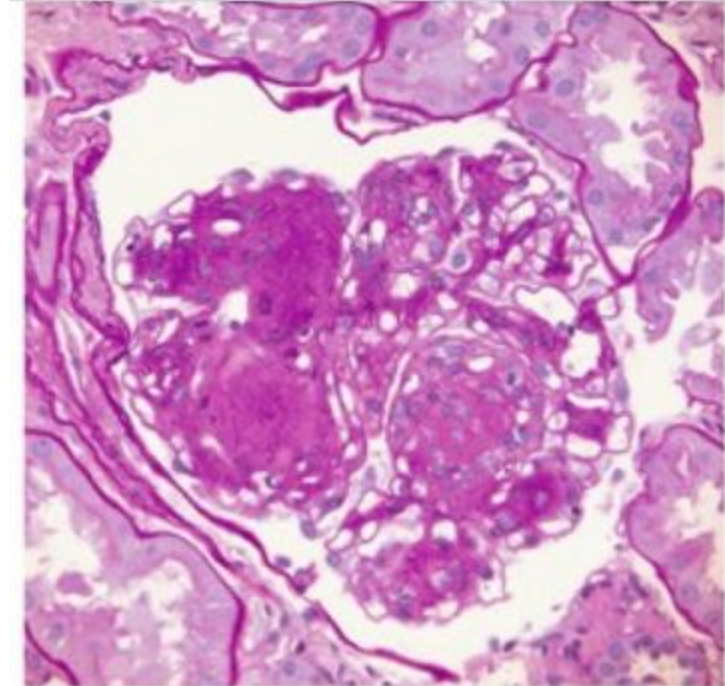
Lim D, Engineering 2022

Normal



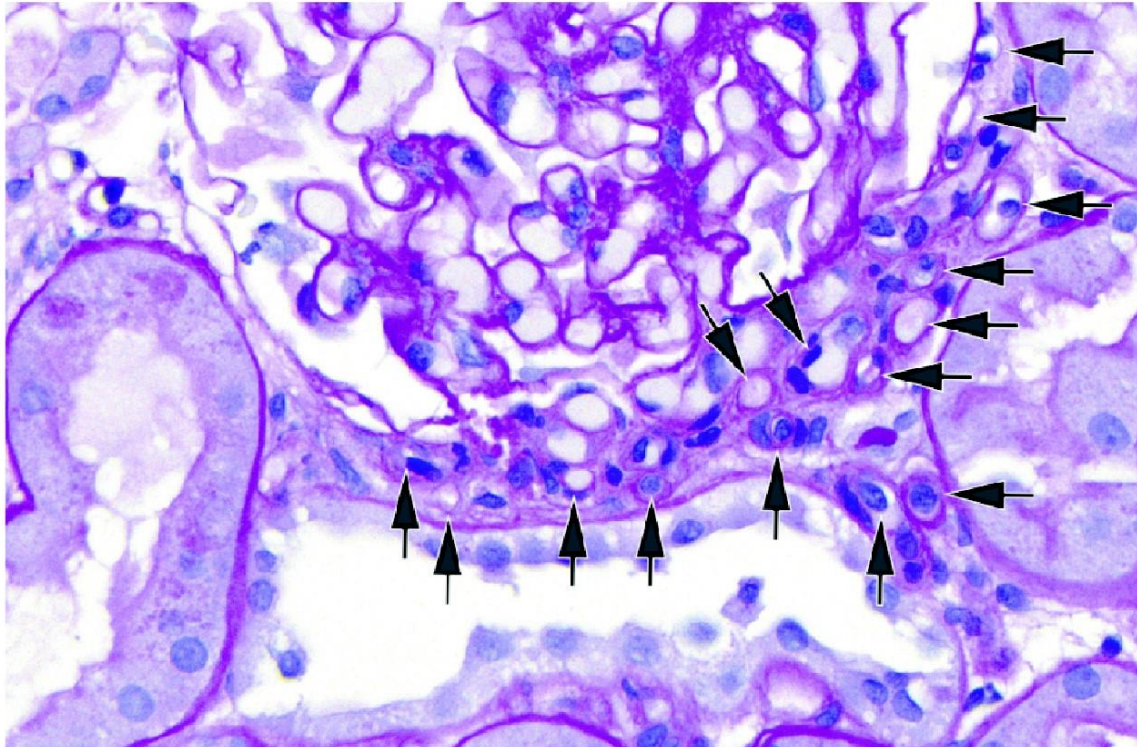
2024, StatPearls Publishing LLC

Diabetic

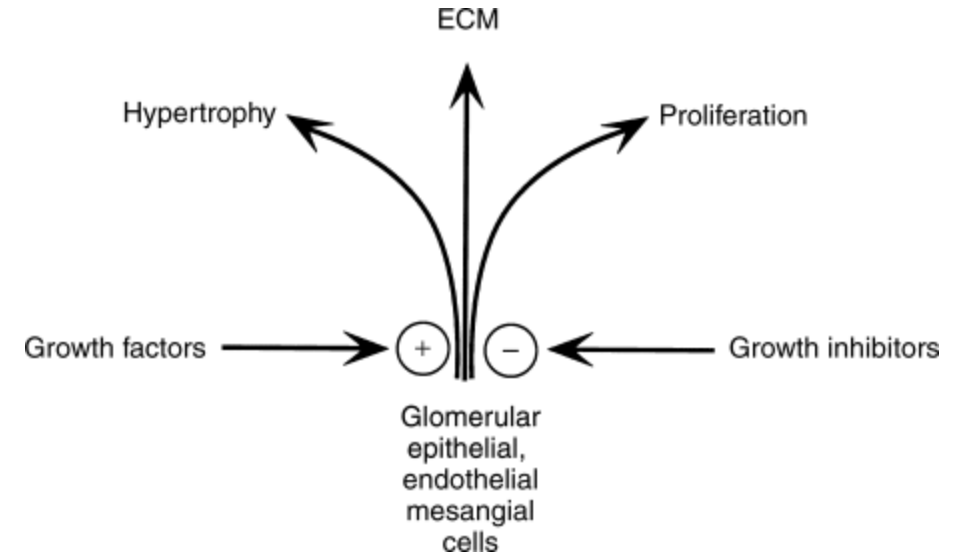


Eadon MT, *Nephrology, Dialysis, Transplantation* 2022

Neovascularization



Eadon MT, *Nephrology, Dialysis, Transplantation* 2022



Fogo A, *Kidney International* 2000

- Neovascularization of afferent and efferent arterioles
- Common in Diabetic Kidney Disease
- Difficult to annotate histologically

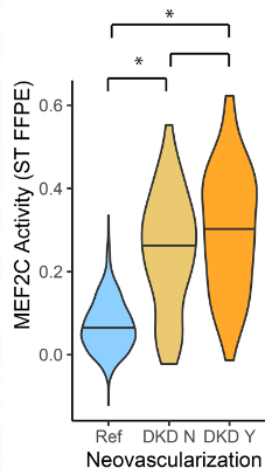
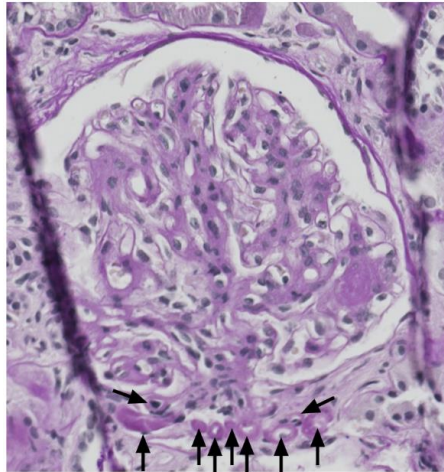
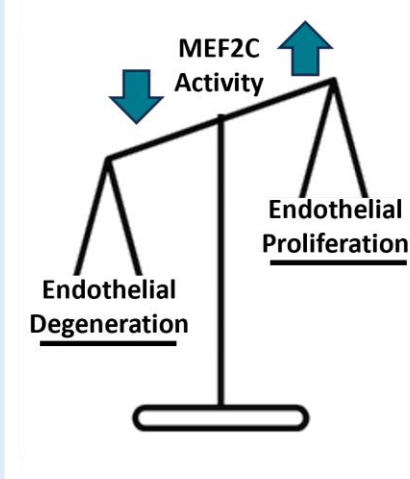
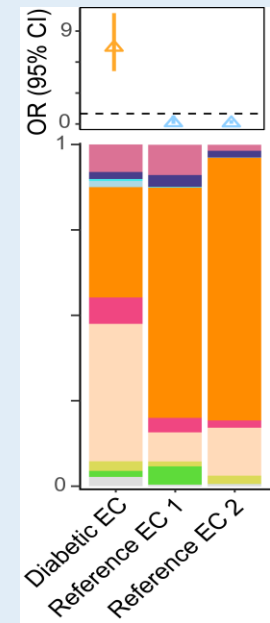
A MEF2C transcription factor network regulates proliferation of glomerular endothelial cells in diabetic kidney disease

Methods and cohort

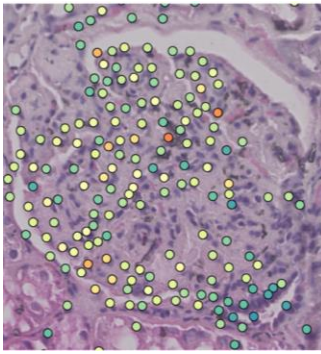
- Performed Multiome or Spatial Transcriptomics in 132 samples from the Kidney Precision Medicine Project and Biopsy Biobank Cohort of Indiana.
- Defined cell state trajectories and Ligand-Receptor interactions by Multiome sequencing.
- Annotated 503 glomeruli for neovascularization and proliferative endothelial cell abundance.
- Assessed MEF2C network changes in a cohort treated with Sodium Glucose Cotransporter 2 inhibitors (SGLT2i).

Findings

Niches of proliferative endothelial cells are specific to diabetic glomeruli.

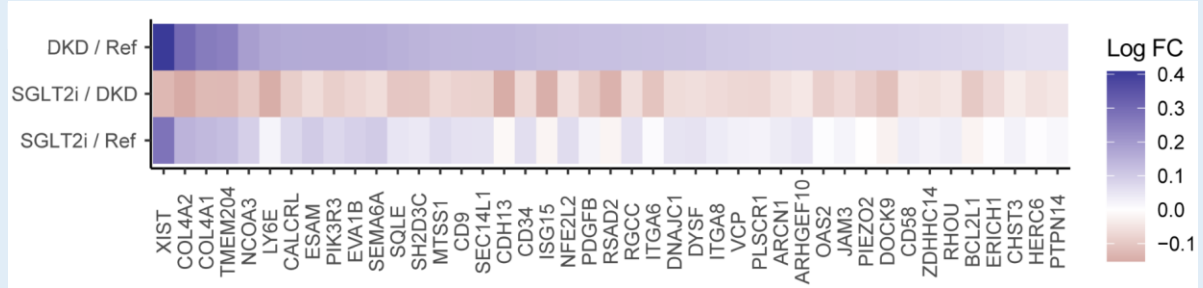


Single cell spatial transcriptomics of inferred MEF2C activity



MEF2C regulates endothelial proliferation + degeneration, localizing to glomerular neovascularization.

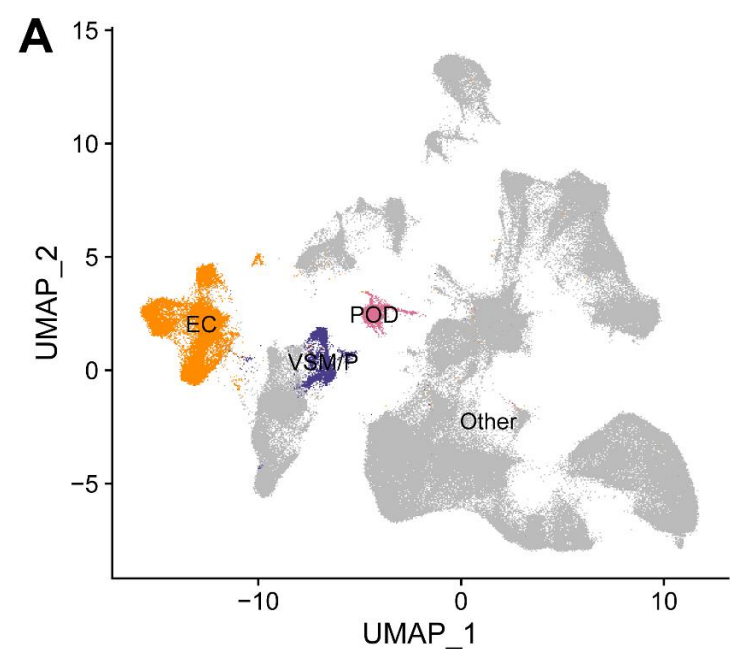
SGLT2i reverses MEF2C network associated expression changes in endothelial cells.



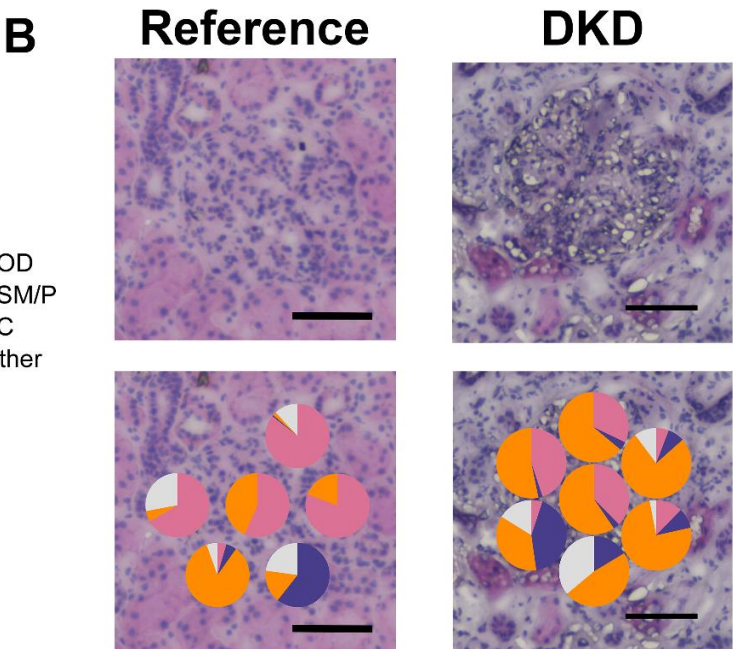
Melo Ferreira et al.,BioRxiv 2025

CONCLUSION The MEF2C, MEF2A, and TRPS1 network regulates proliferation and degeneration of glomerular capillary endothelial cells in Diabetic Kidney Disease. Sodium Glucose Cotransporter 2 inhibitor therapy reversed the MEF2C network expression changes seen in diabetic kidney disease.

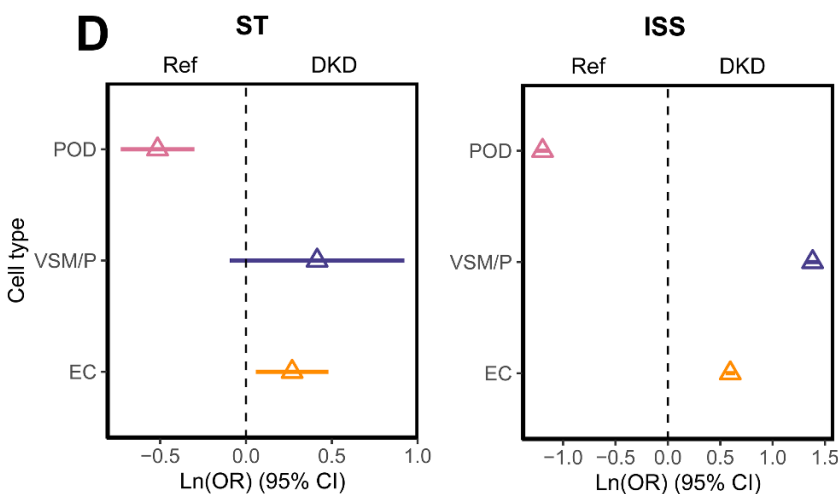
Endothelial cell hypercellularity in DKD



Subset of potential glomerular cells

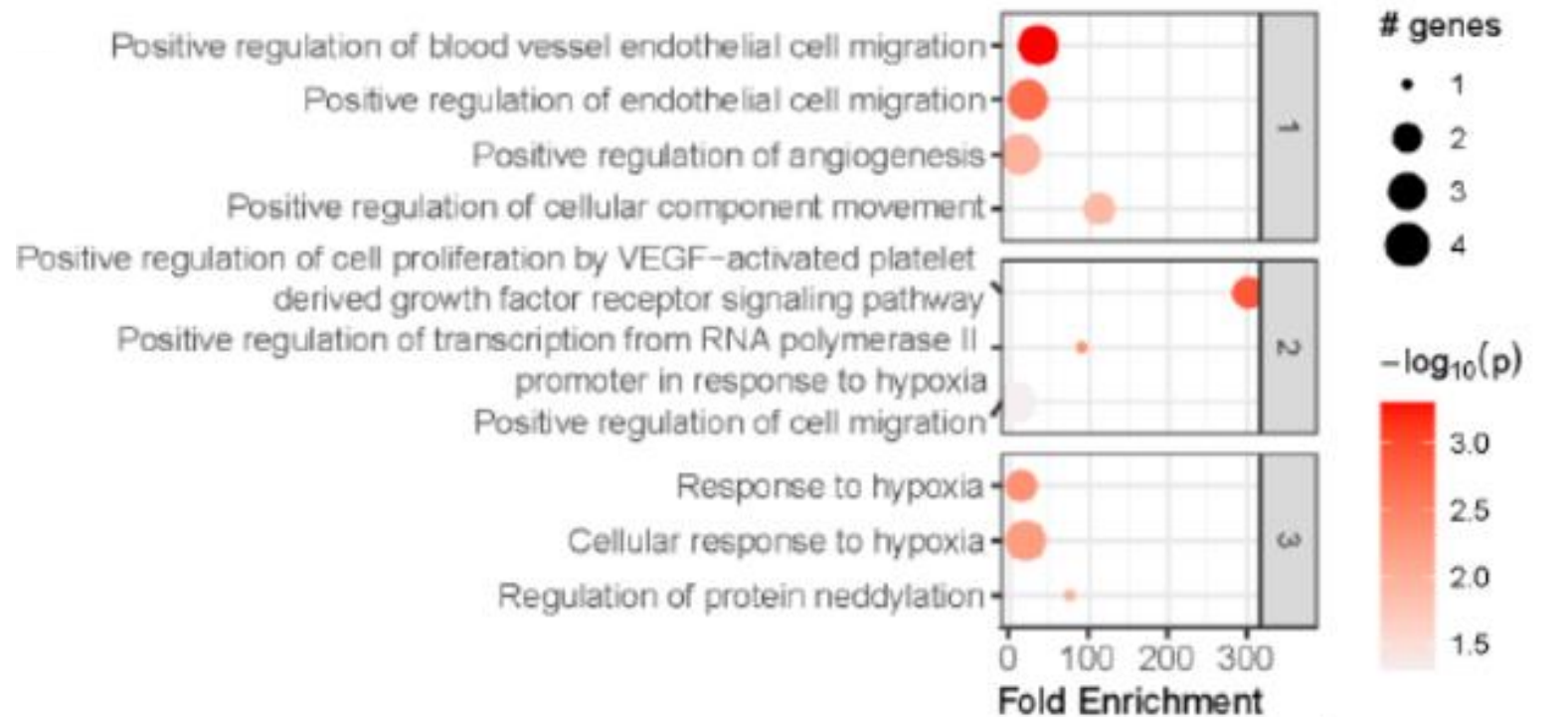
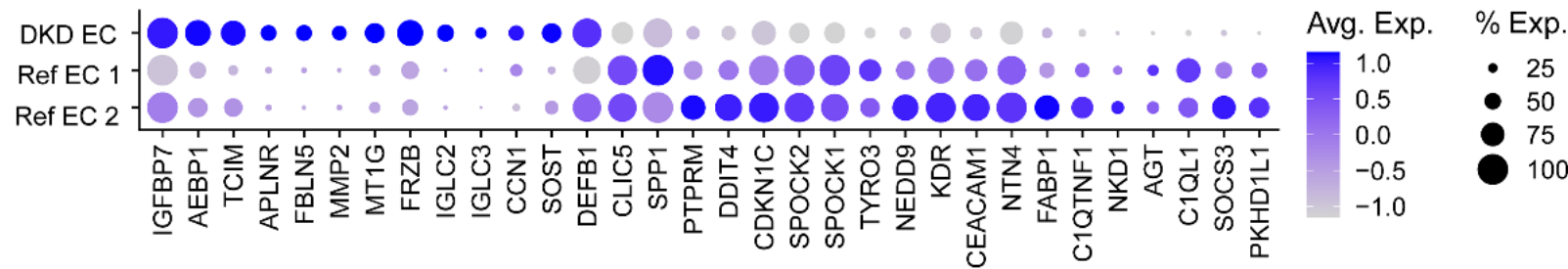
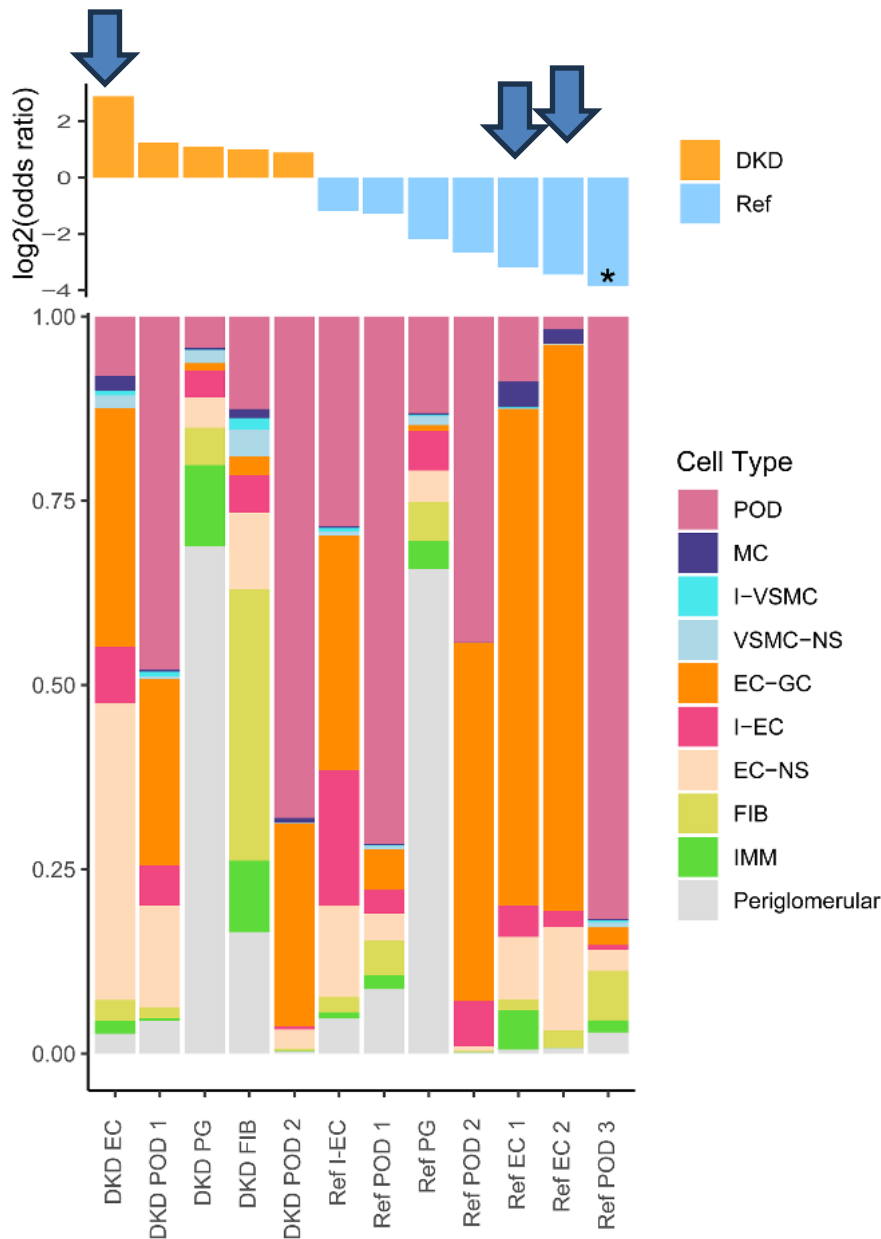


Label transfer suggests difference in glomerular cell type composition

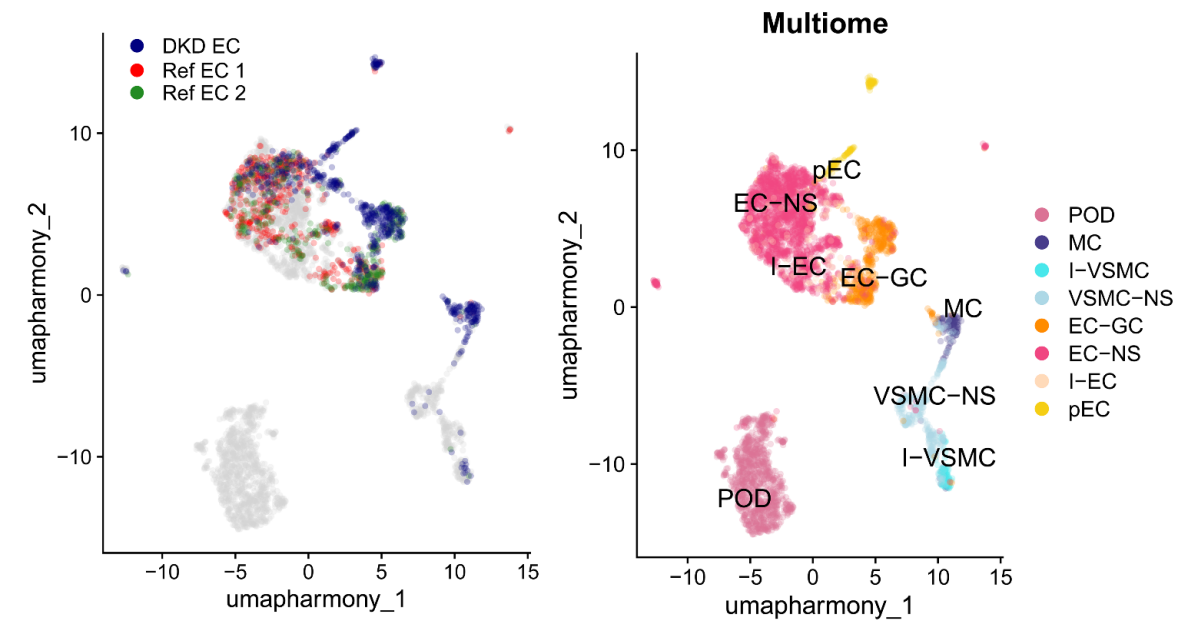


Reduction in podocyte signature. Increase in VSM/P + EC.

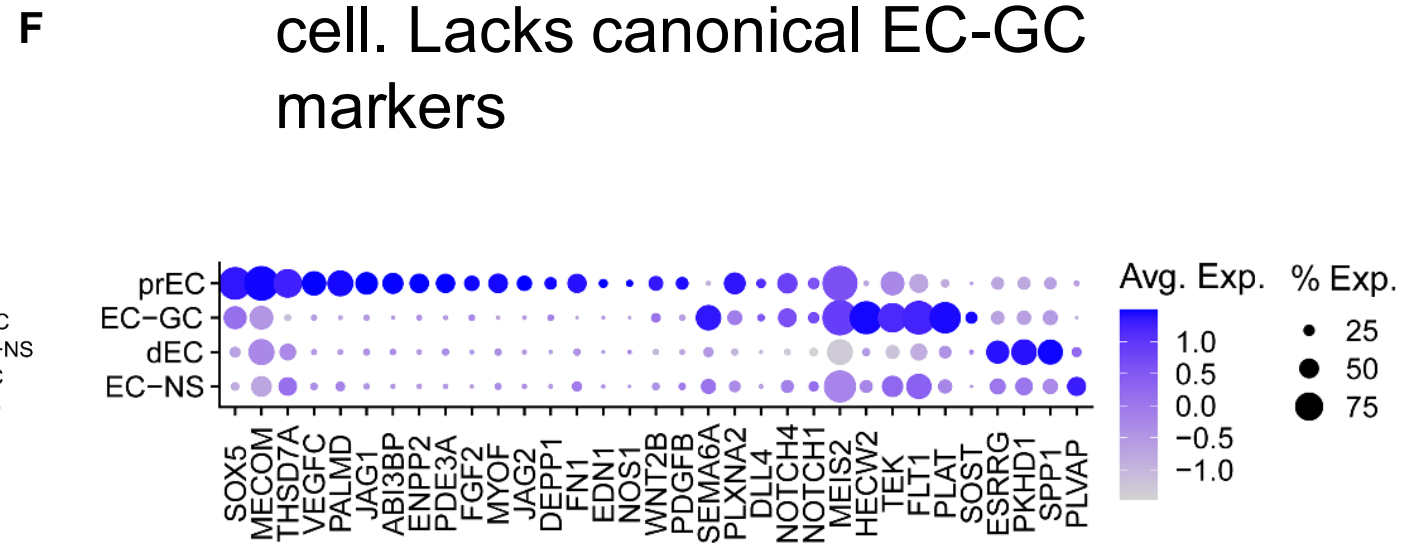
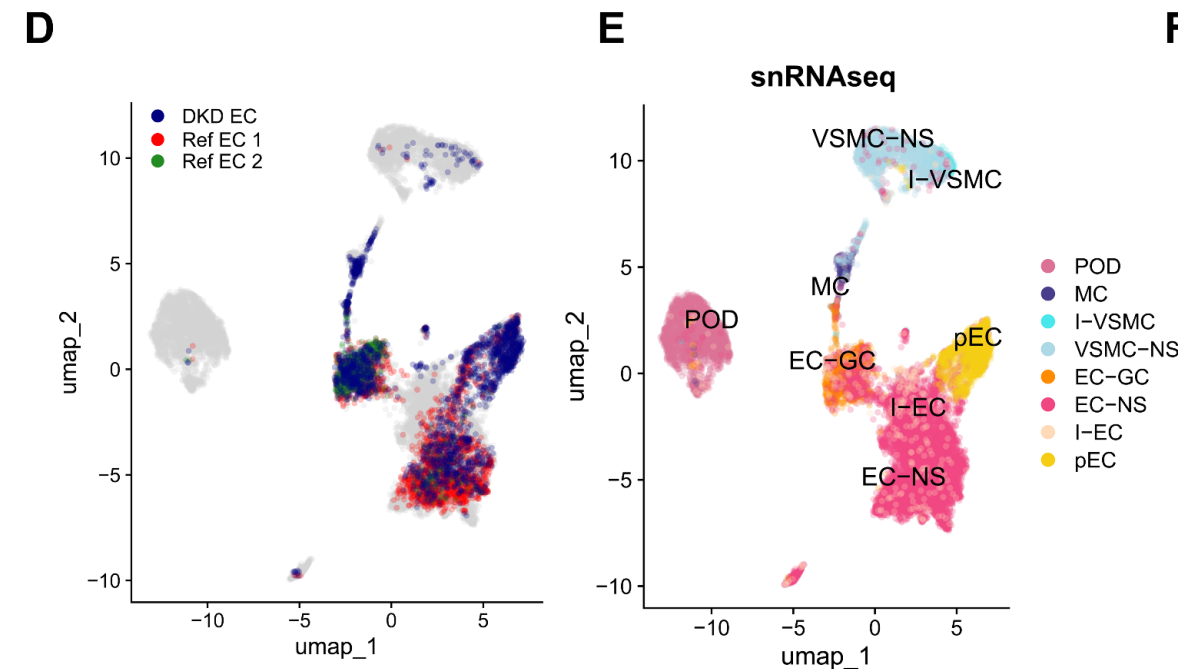
Neighborhood analysis



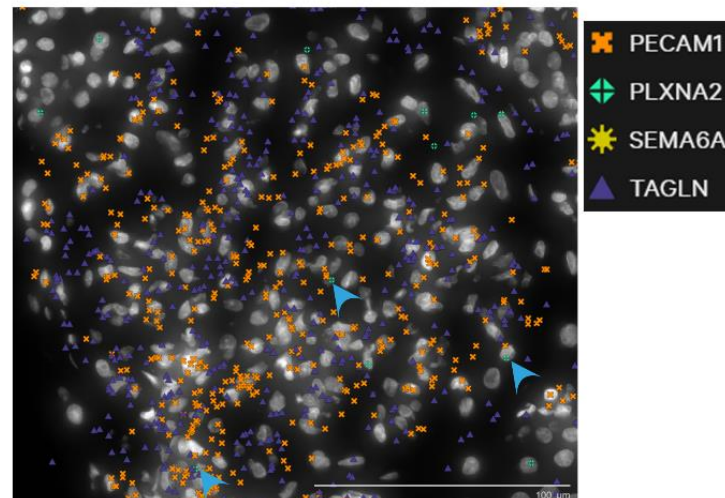
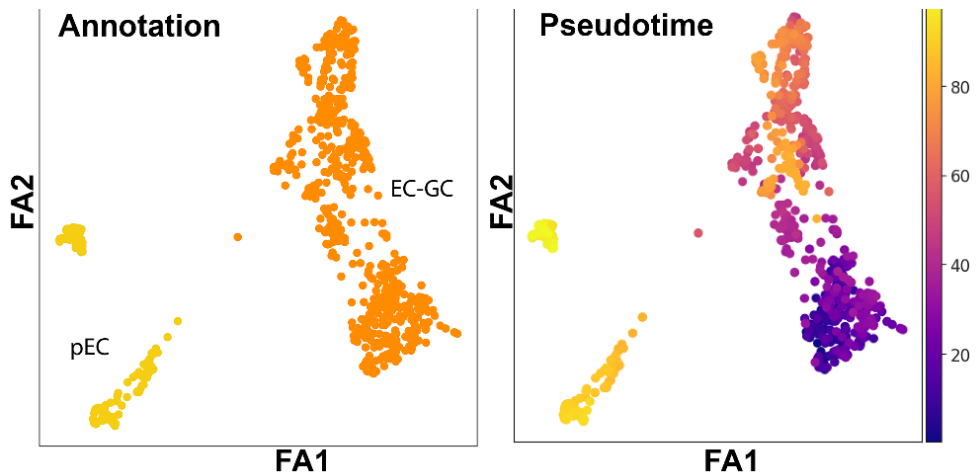
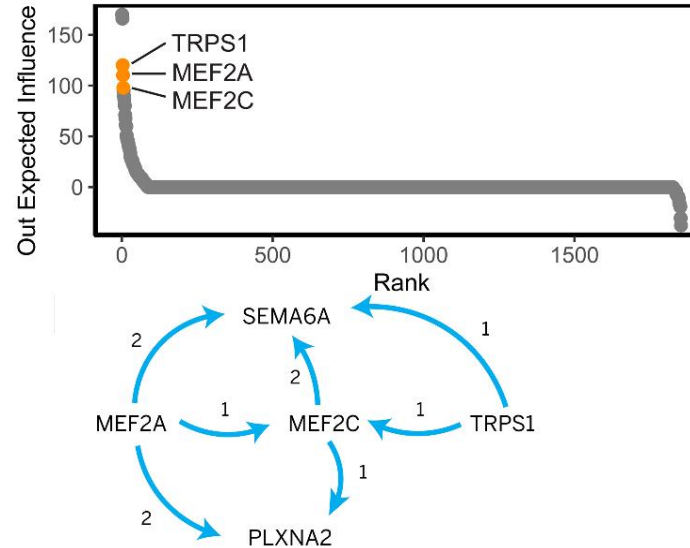
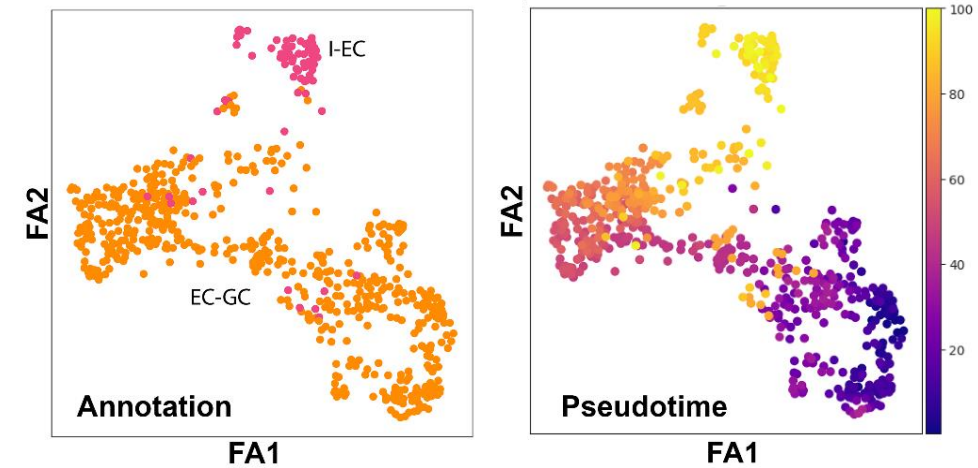
DKD Niche pathways enriched for Hypoxia and Angiogenesis



- Back-mapped ST neighborhood information (reverse label transfer)
- Determined which endothelial sub-cluster is in diabetic glomeruli.
- prEC for proliferative endothelial cell. Lacks canonical EC-GC markers



Trajectories of glomerular endothelial cell to degenerative or proliferative phenotypes

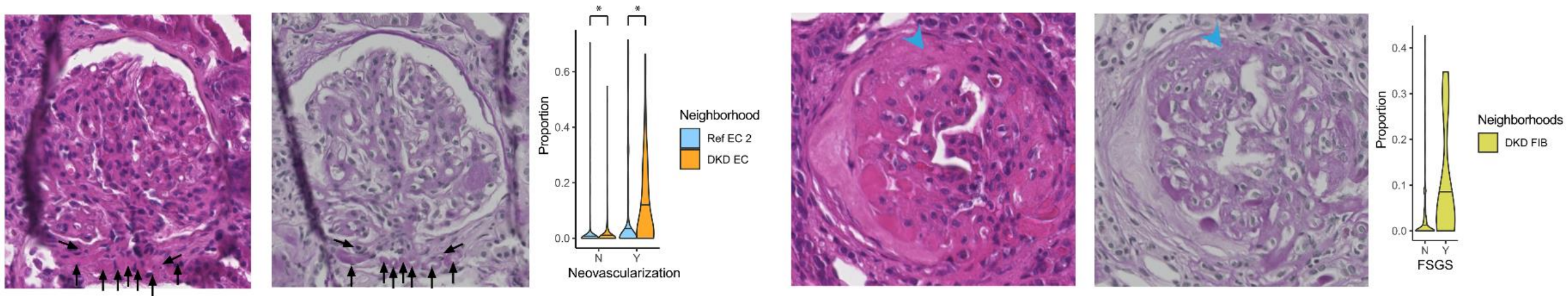


Then we did this:

- Defined TF networks governing trajectories toward degenerative or proliferative endothelial cells
- Defined receptor ligands
- Spatially validated R-L pairs at single cell level
- In silico knockout of TFs
- In vitro knockout of TFs in human EC-GCs
- ChIP-seq for binding validation
- Protein Validation

Neovascularization lesions correlate with prEC niches

FSGS lesions correlate with Fibroblast niches

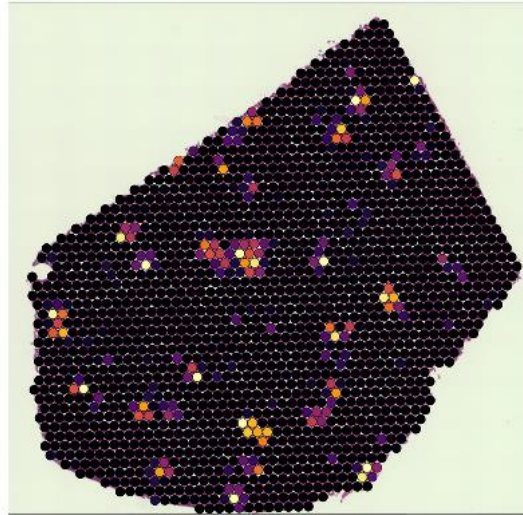


- Scored every glomerulus based on histology
- Molecularly defined neighborhoods align with expected histology

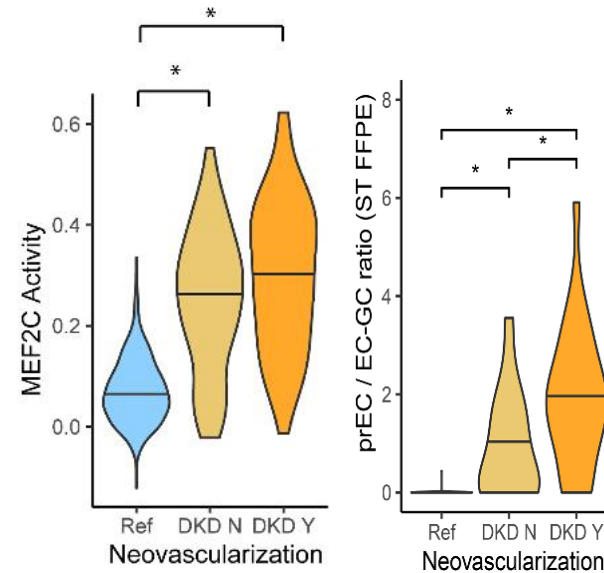
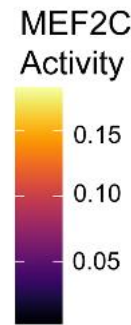
MEF2C activity mapping

MEF2C activity increased in DKD and upregulated further in nodular glomeruli.

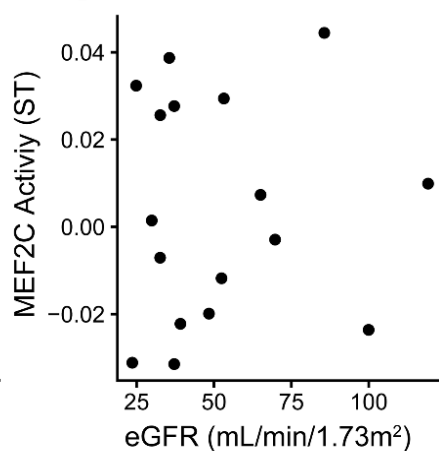
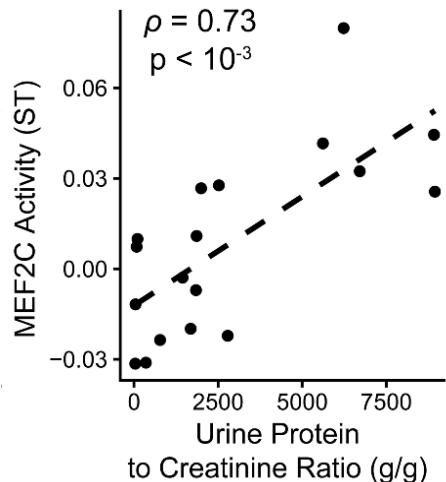
Reference



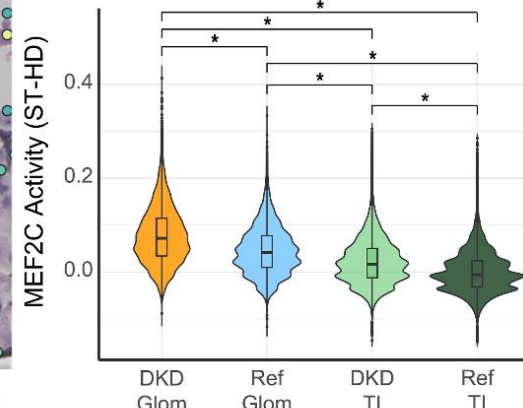
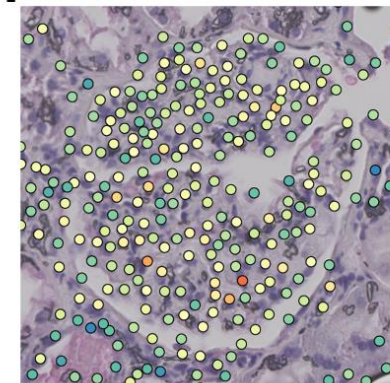
DKD



Inferred MEF2C activity and prEC abundance mapped to glomeruli with neovascularization

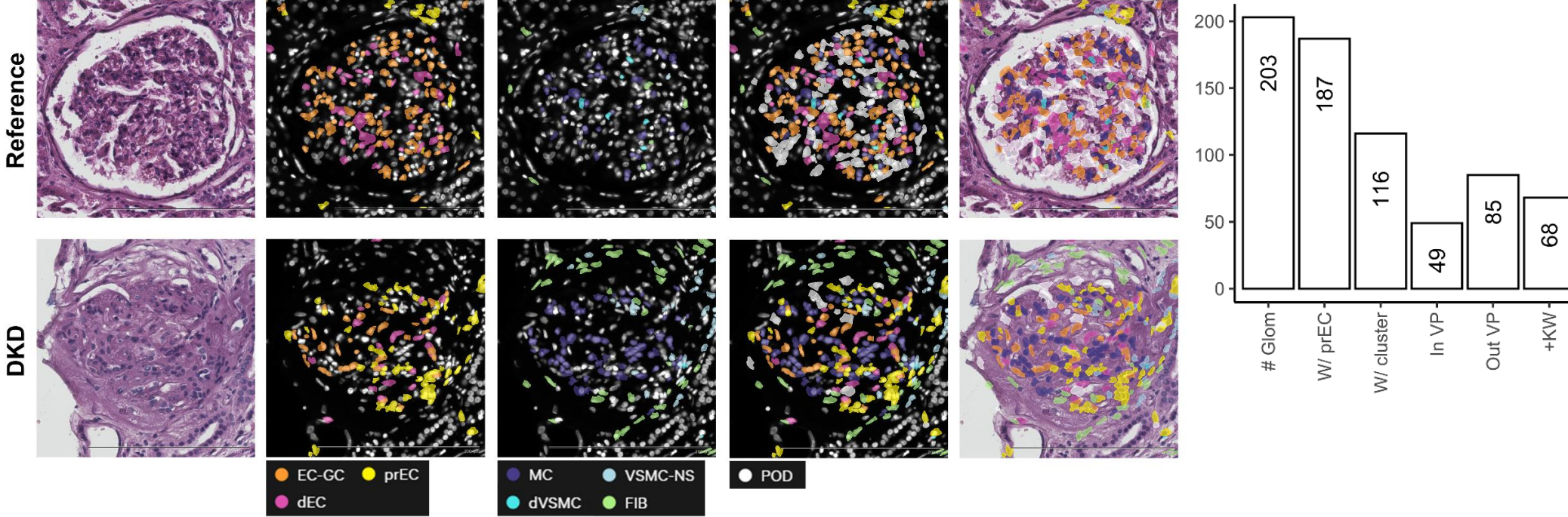


MEF2C activity correlated with proteinuria but not eGFR in the KPMP

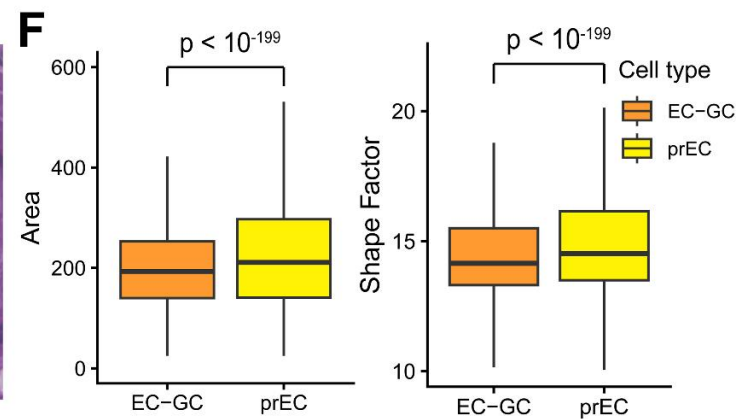
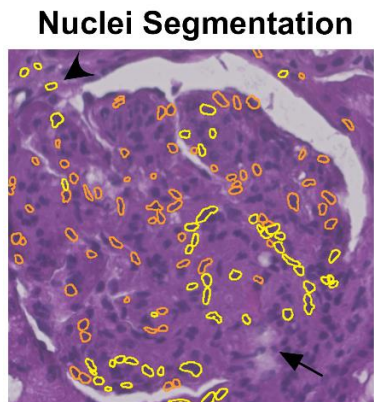


Inferred MEF2C activity is highest in DKD Glom

Spatial changes in prEC + VSMC cells

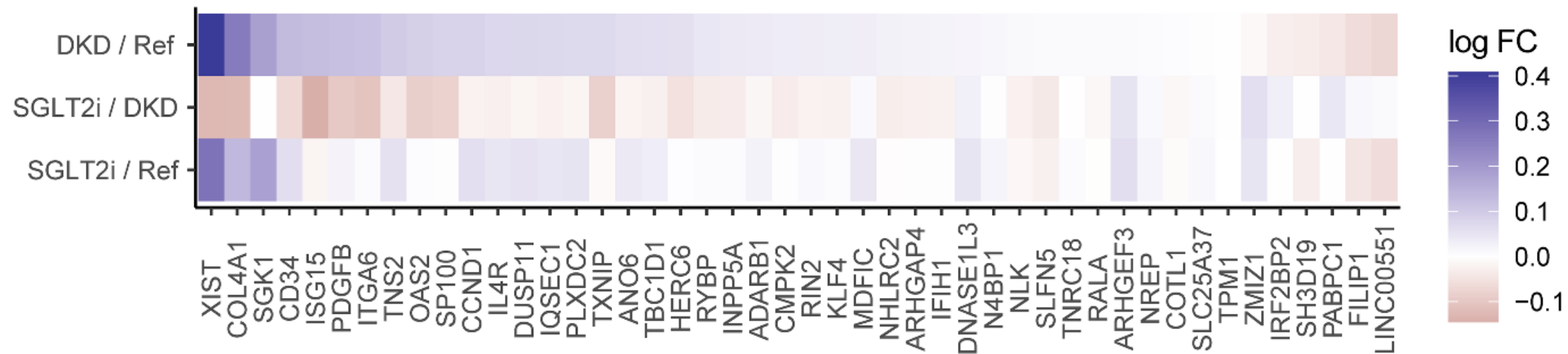


prECs are broadly distributed, but tend to cluster and are found near the vascular pole and adjacent to KW nodules



prEC Nuclei vary in size and shape from canonical EC-GCs

SGLT2i reverses DKD induced changes in the MEF2C network



- *scRNA-seq from research kidney biopsies from T2D aged 12 to 21 years, and healthy controls (Ref).*
- *Ten T2D participants had been prescribed SGLT2i (T2Di[+]) and 6 not (T2Di[-]).*
- *Transcriptional profiles showed MEF2C regulated genes in glomerular capillary endothelial cells..*

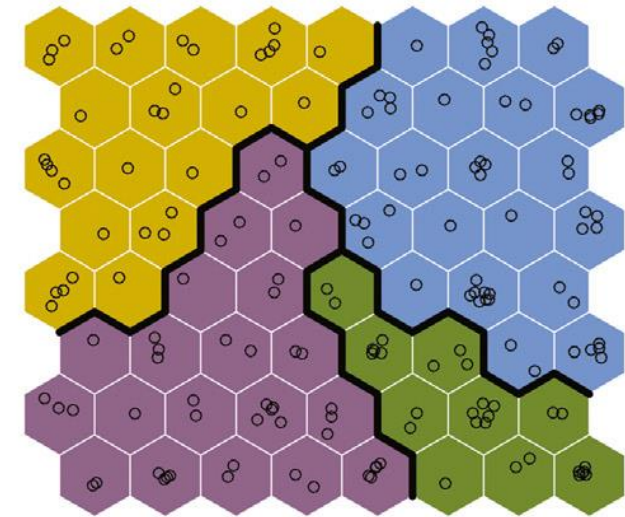
Residual Risk of Diabetic Kidney Disease

Supervised

- Sub-phenotype on biomarkers representative of clinical / histopathologic features
- Examples:
 - Disproportionate Podocytopathy
 - Neovascularization
 - Glomerular hypertension
 - Metabolic profile of epithelial cells
 - Nerve / Renal Tubular Acidosis
 - Inflammation: B cell, eosinophil, mast cell, etc
 - Concentration, Acid-base handling
 - Salt avidity

Unbiased

- Cluster patients based on molecular profile to identify those with residual risk factors.
- Example:
 - Self-organizing map to drive grouping of patients.
 - We sought to update this for the KPMP with multi-cellular programs (analogous to Austin criteria)

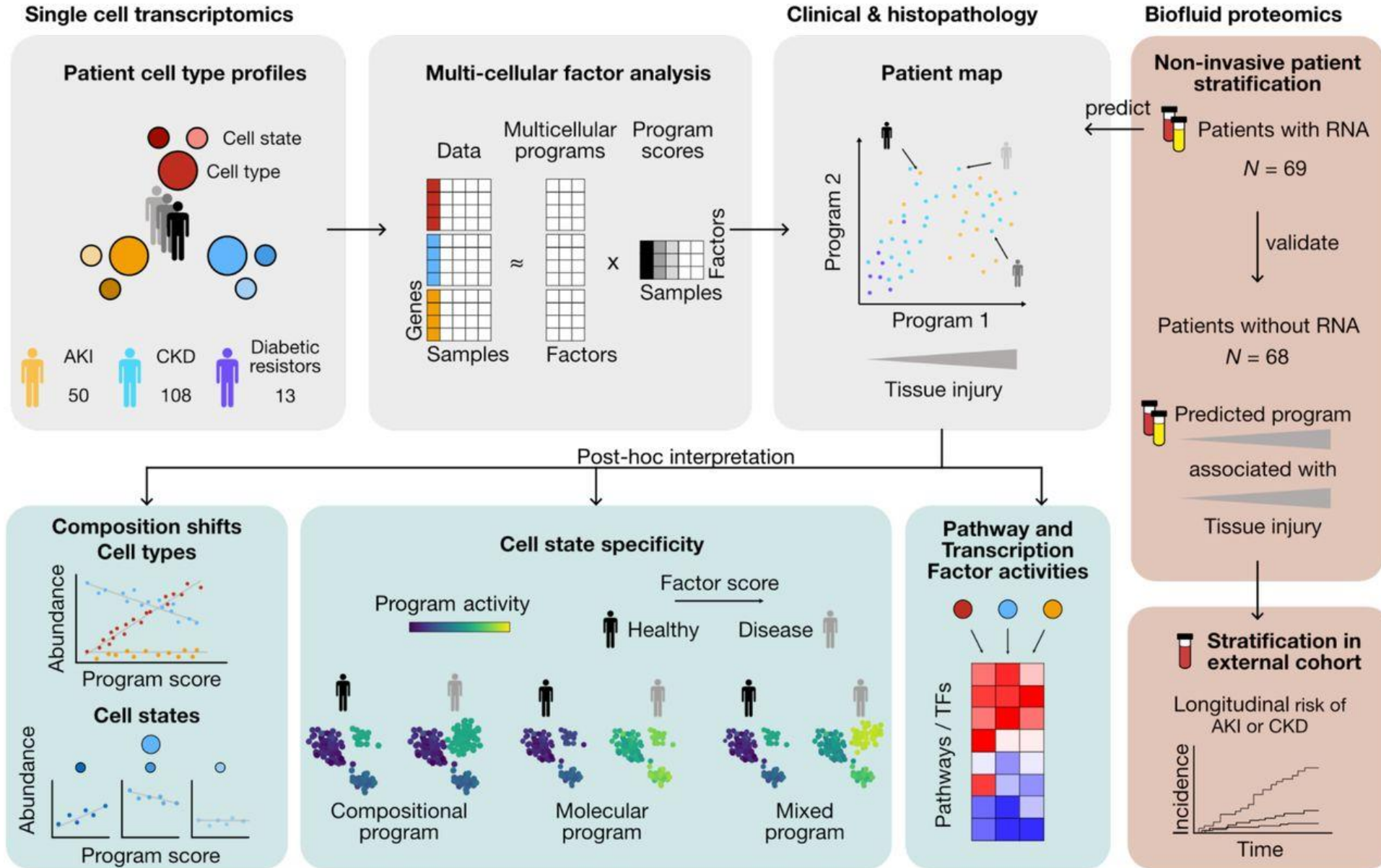


Reznichenko A, KI 2024

Multicellular Factor Analysis (snRNA-seq)



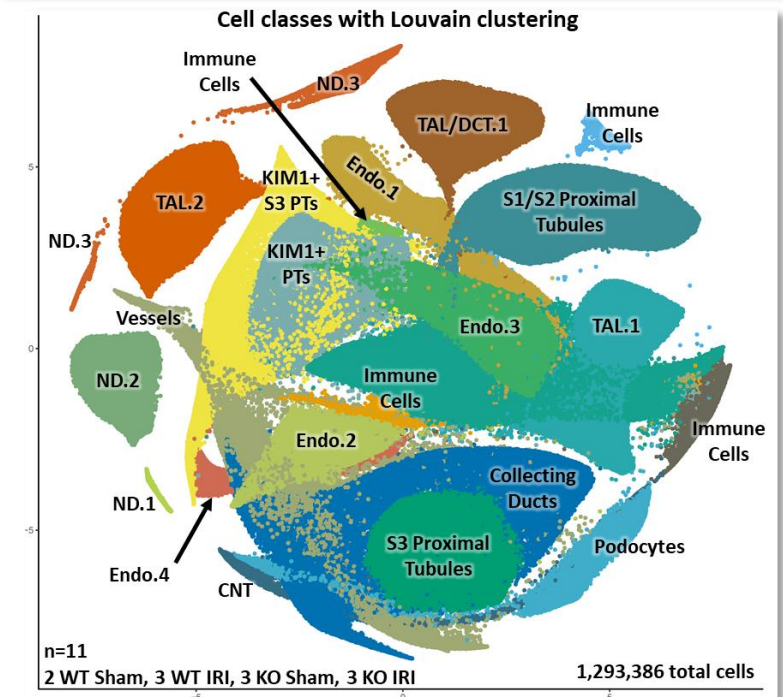
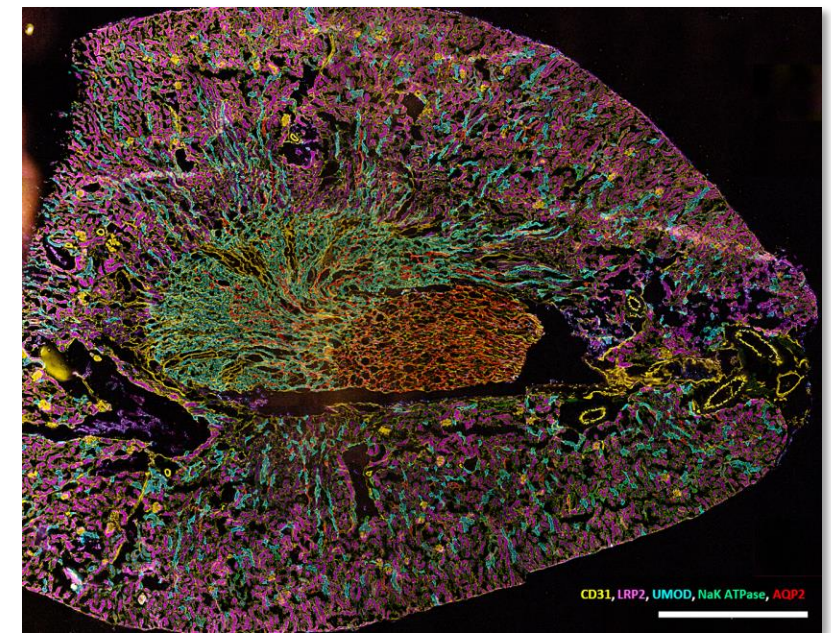
Robin Fallegger





Summary

- Many groups that analyze snRNA-seq
- We start with the spatial!
- We use niches, orthogonal readout adjacency, histopathology of functional tissue units to interpret how cells change in disease.
- Examples:
 - Trajectories of epithelial regeneration (THY1)
 - Trajectories that balance maladaptation + recovery (ELF3, SOX4)
 - Trajectories balancing endothelial proliferation + degeneration (MEF2C)
- Important work underway for unbiased clustering of spatial molecular and protein data



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3D Tissue Imaging Core**

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